

BFS Traversal

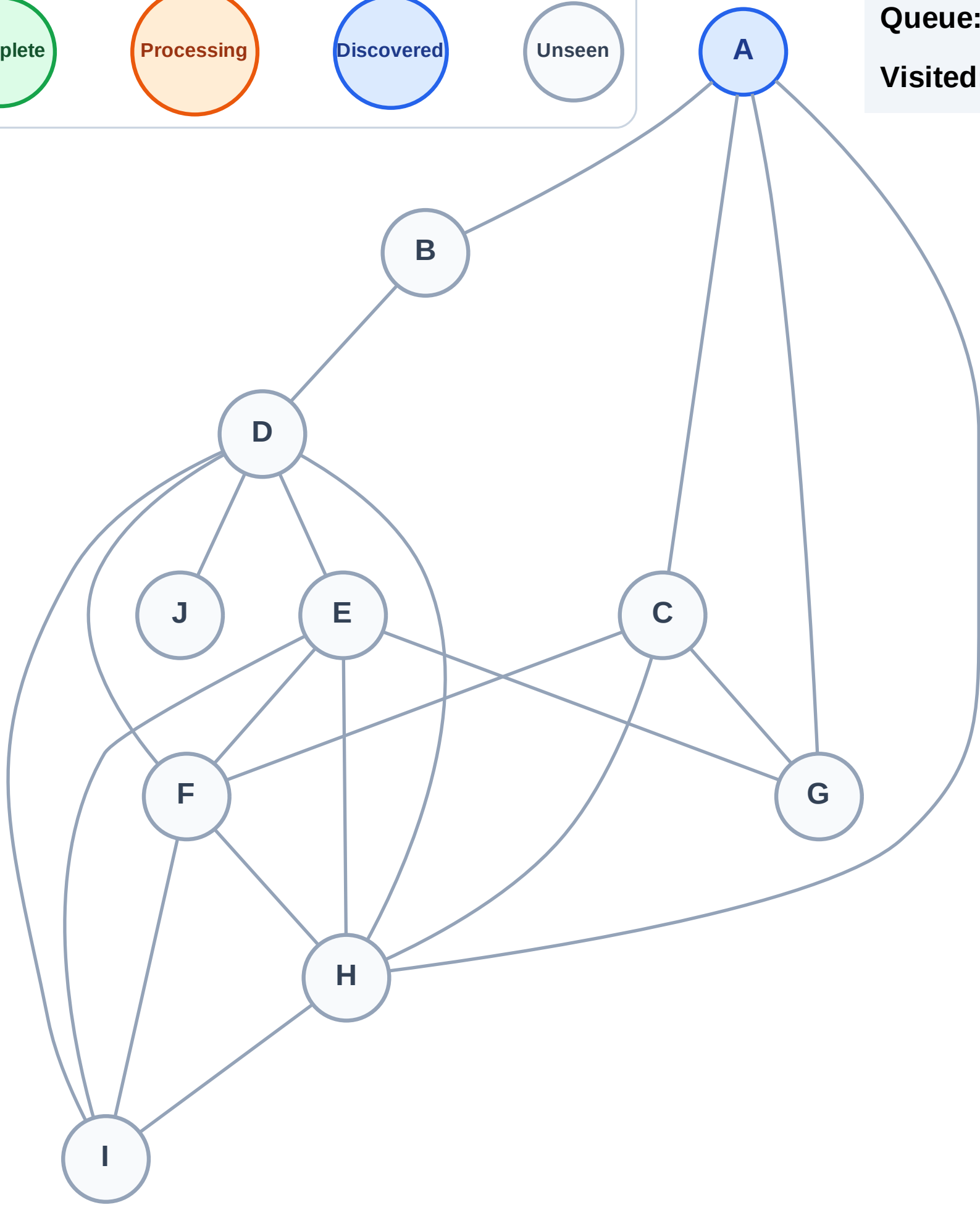
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

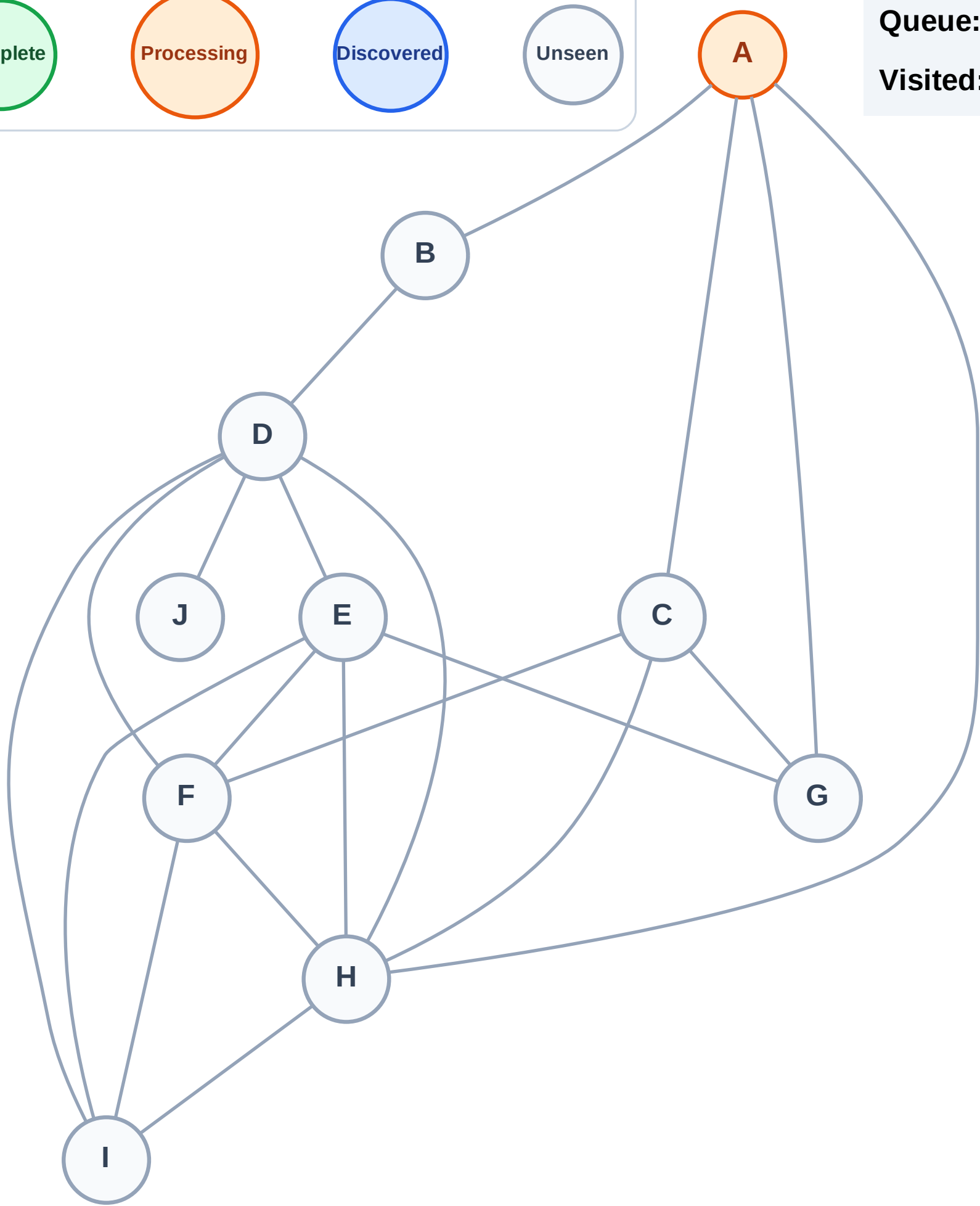
Step 2: Process node A

Legend



Queue: (empty)

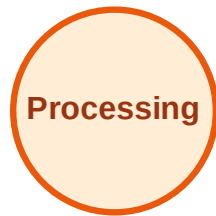
Visited: (none)



BFS Traversal

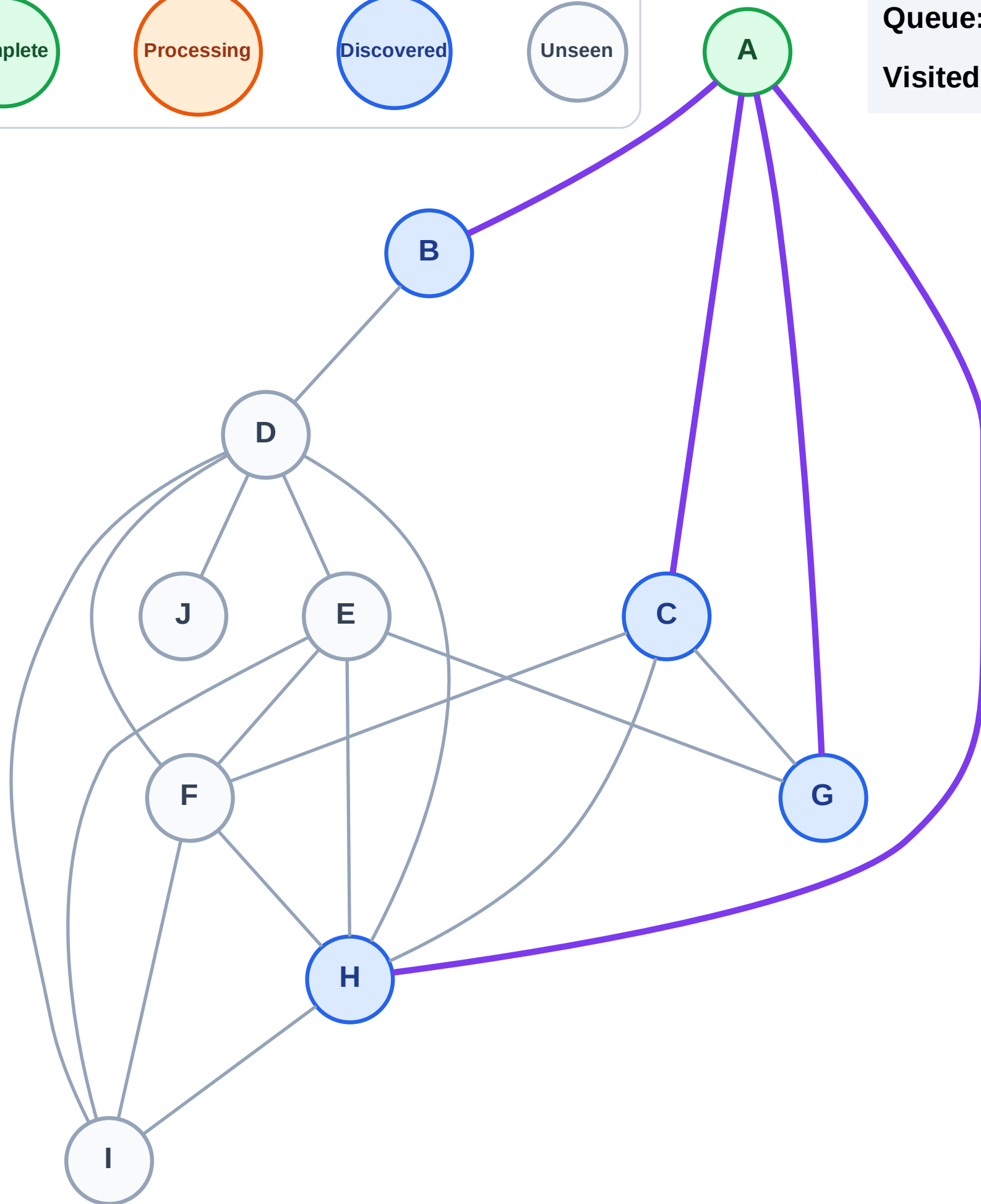
Step 3: Discover B, C, G, H from A

Legend



Queue: B → C → G → H

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

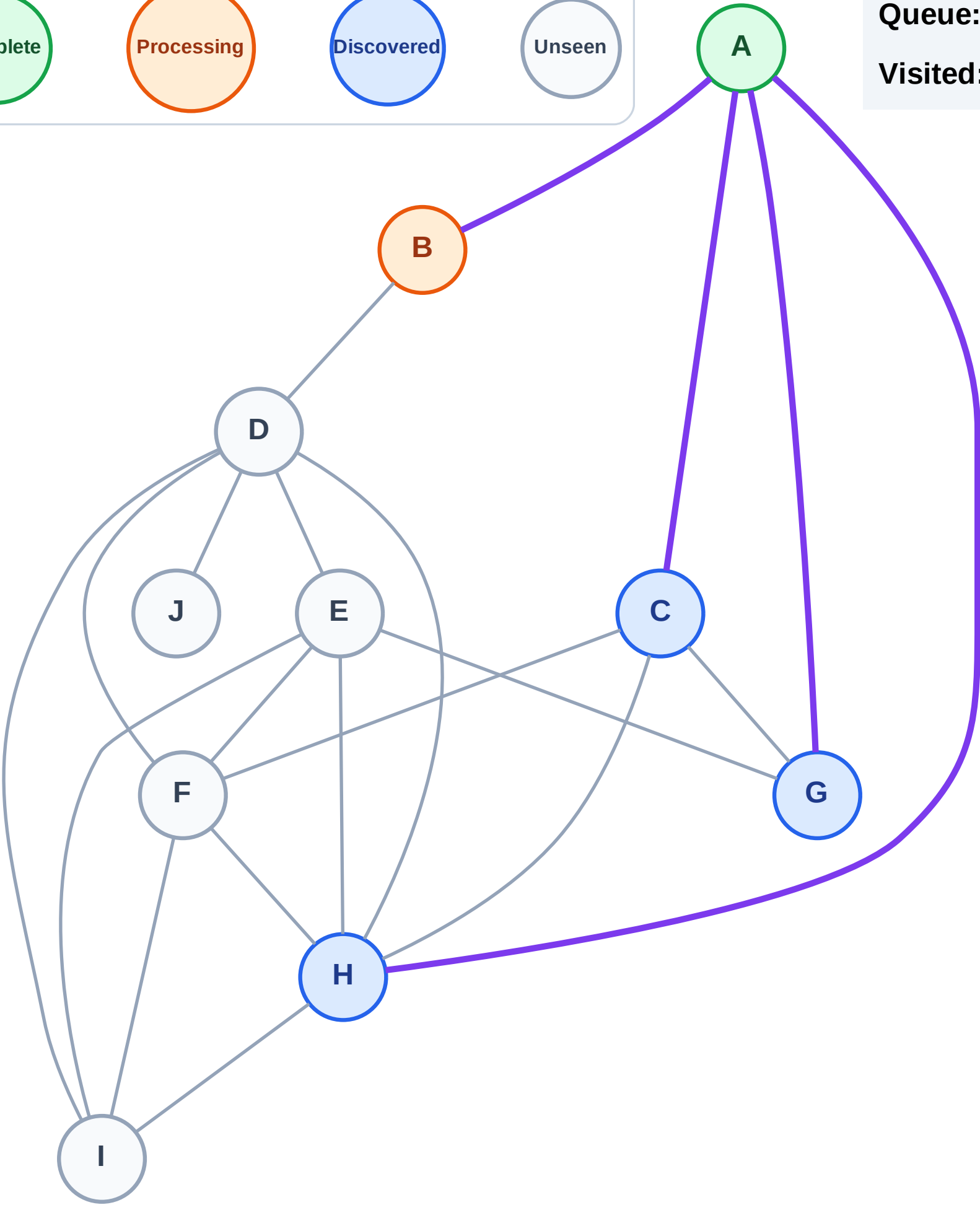
Processing

Discovered

Unseen

Queue: C → G → H

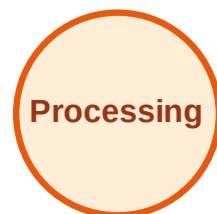
Visited: A



BFS Traversal

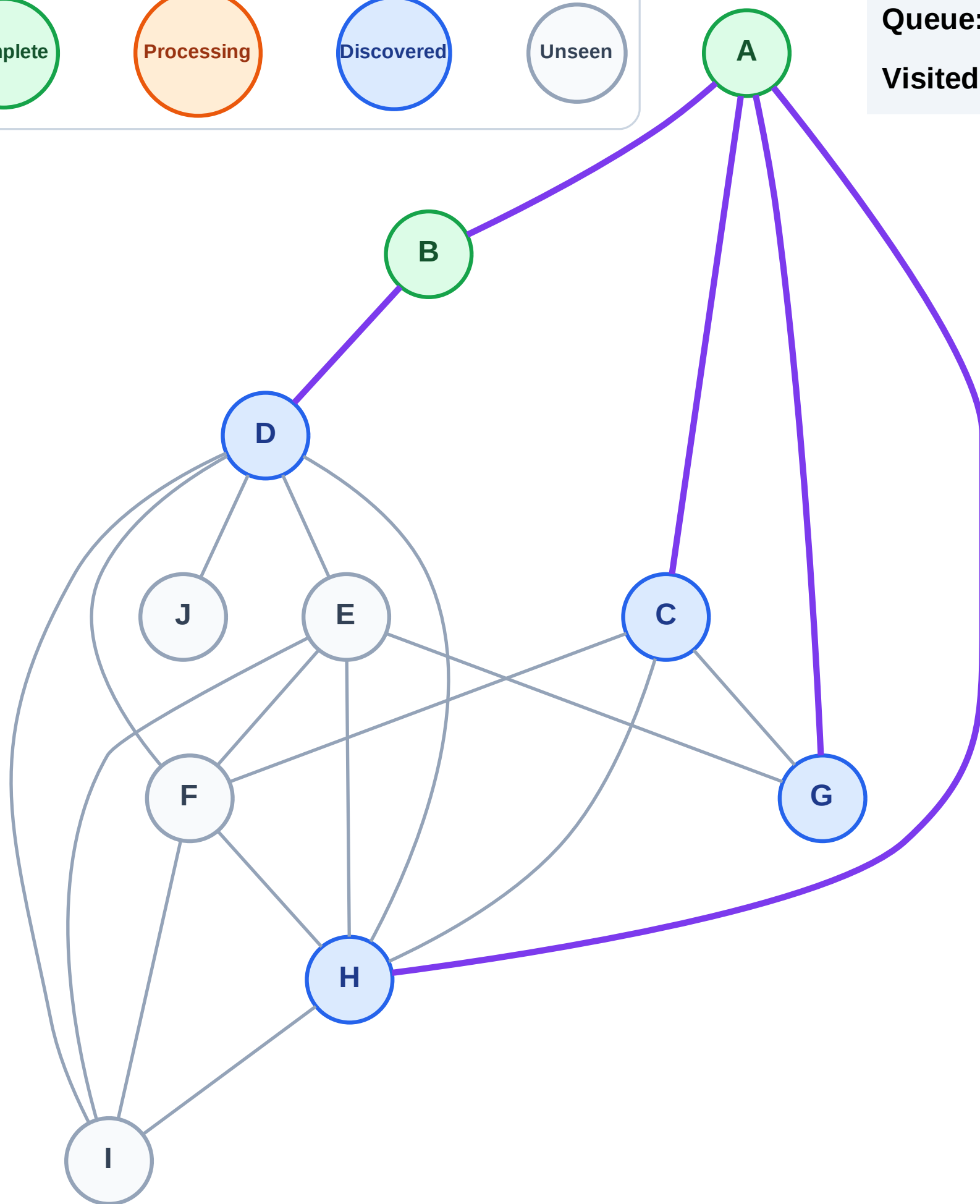
Step 5: Discover D from B

Legend



Queue: C → G → H → D

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

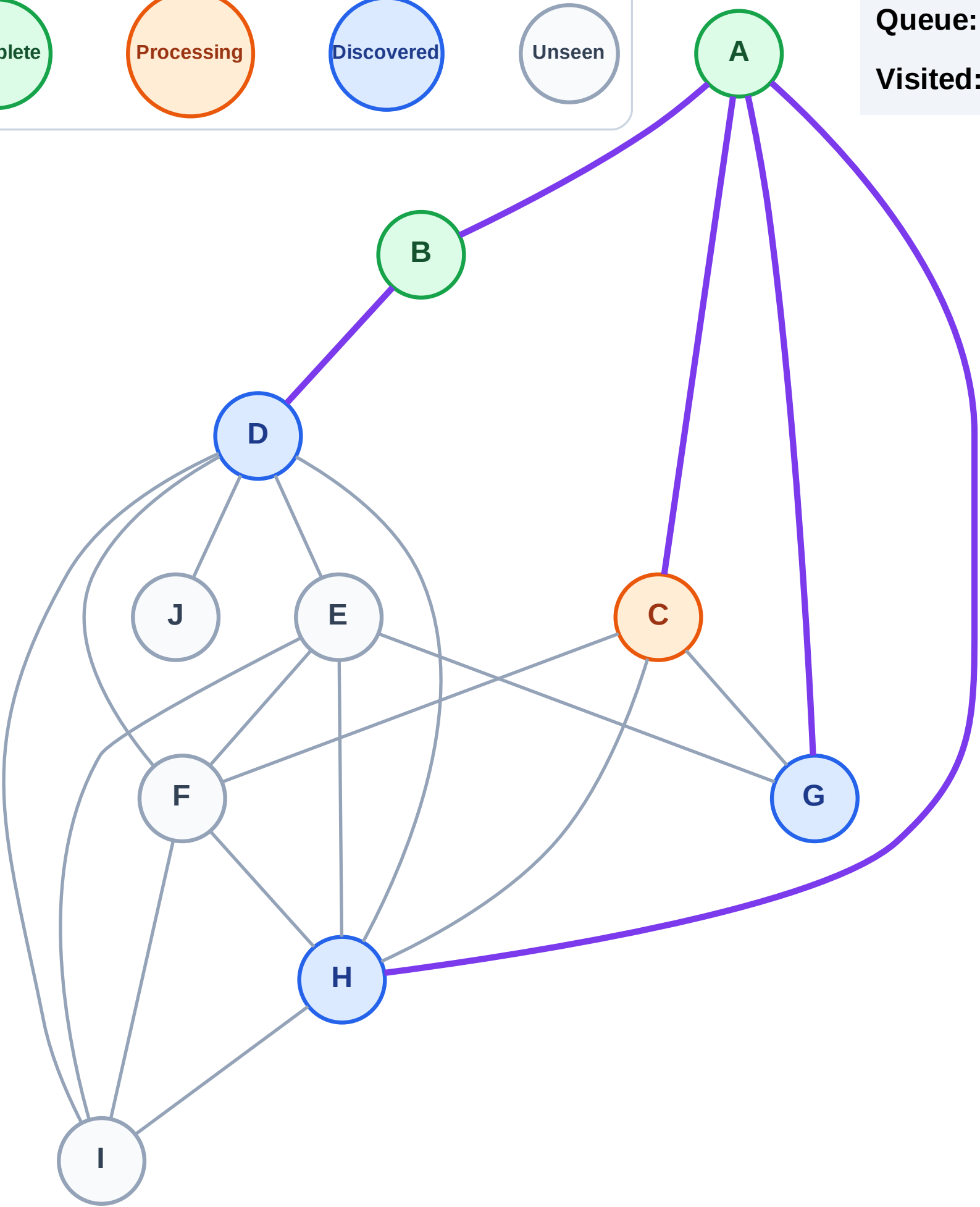
Complete

Processing

Discovered

Unseen

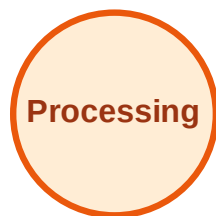
Queue: G → H → D
Visited: A, B



BFS Traversal

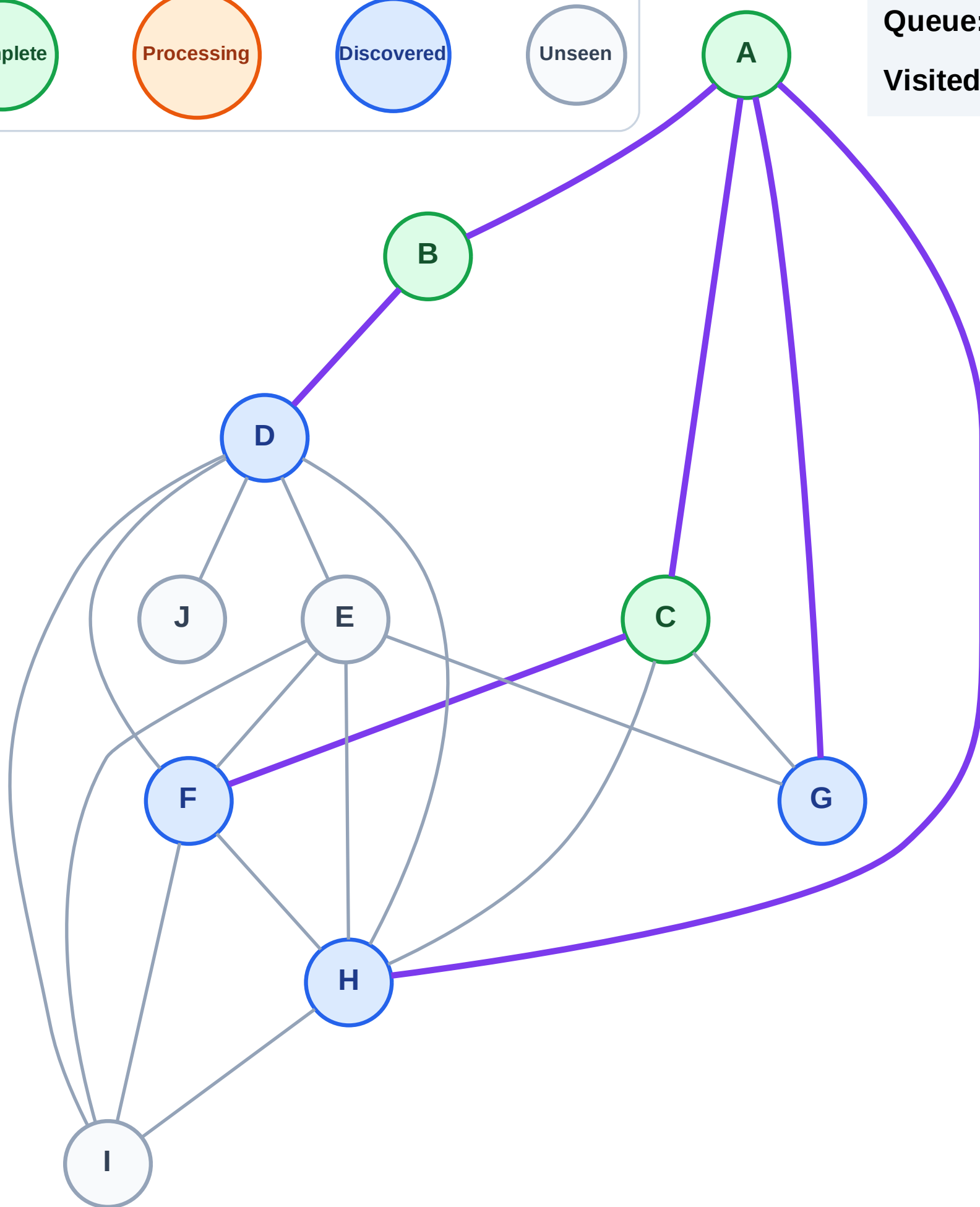
Step 7: Discover F from C

Legend



Queue: G → H → D → F

Visited: A, B, C



BFS Traversal

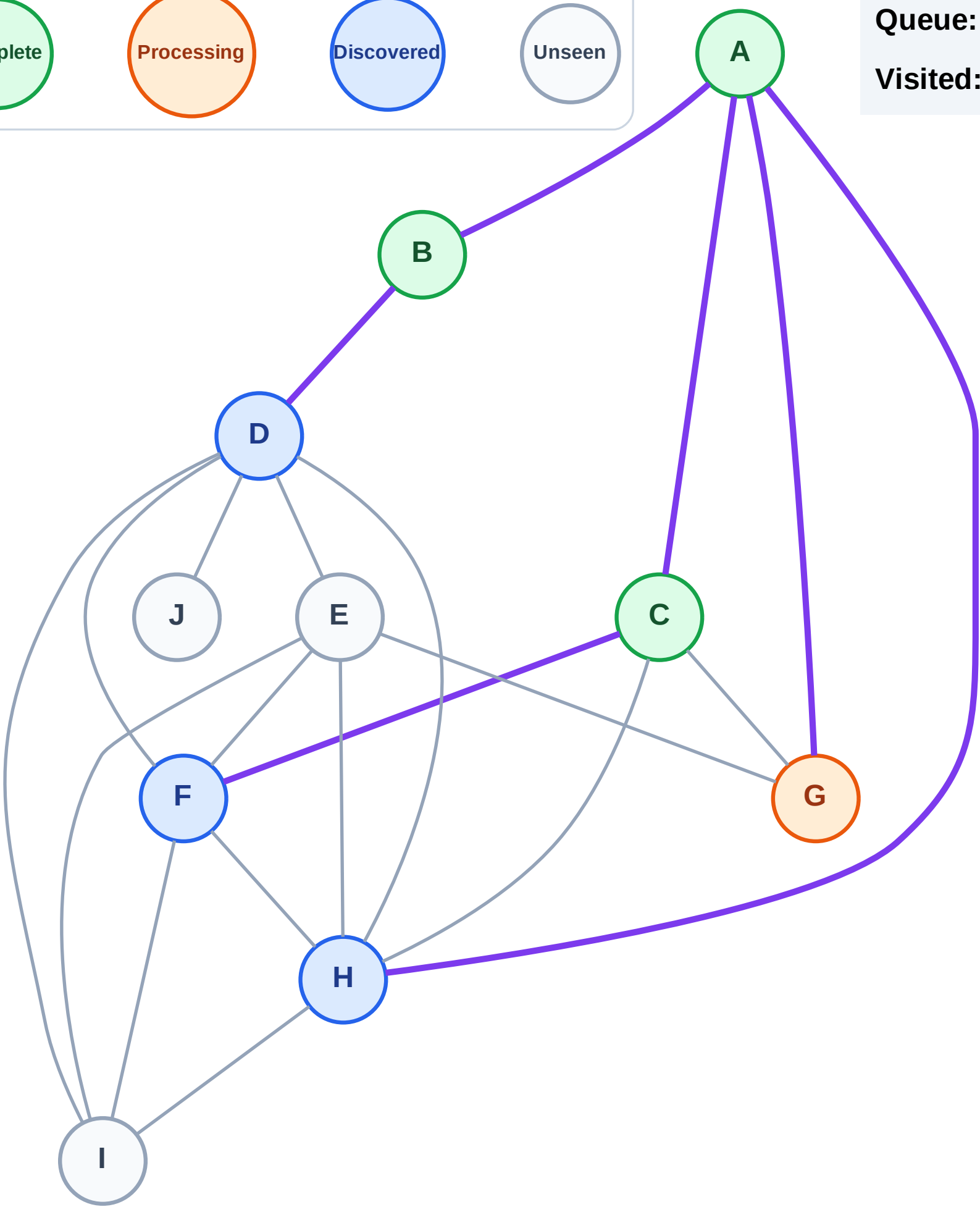
Step 8: Process node G

Legend



Queue: H → D → F

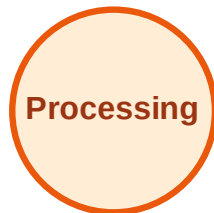
Visited: A, B, C



BFS Traversal

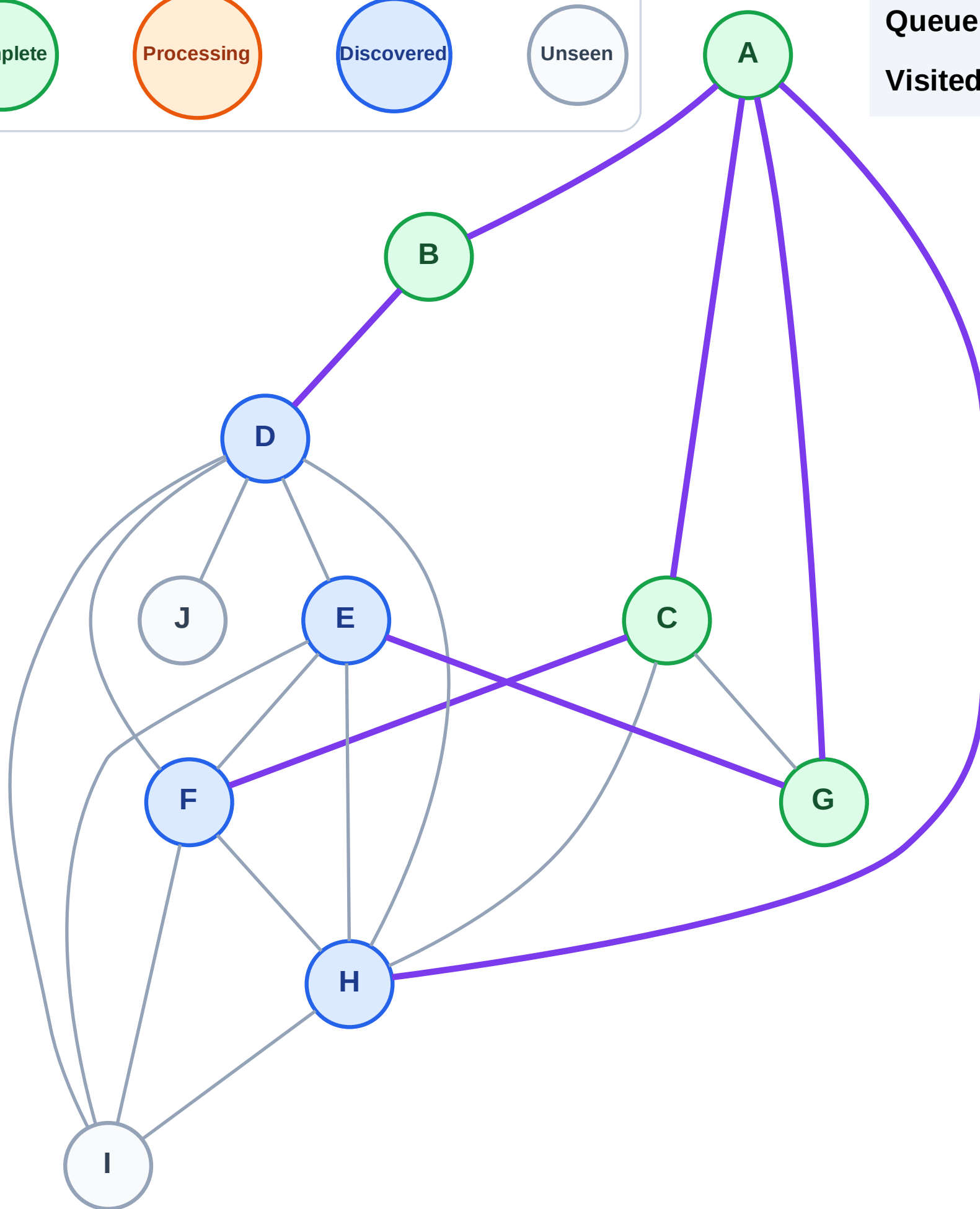
Step 9: Discover E from G

Legend



Queue: H → D → F → E

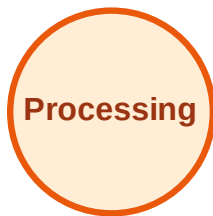
Visited: A, B, C, G



BFS Traversal

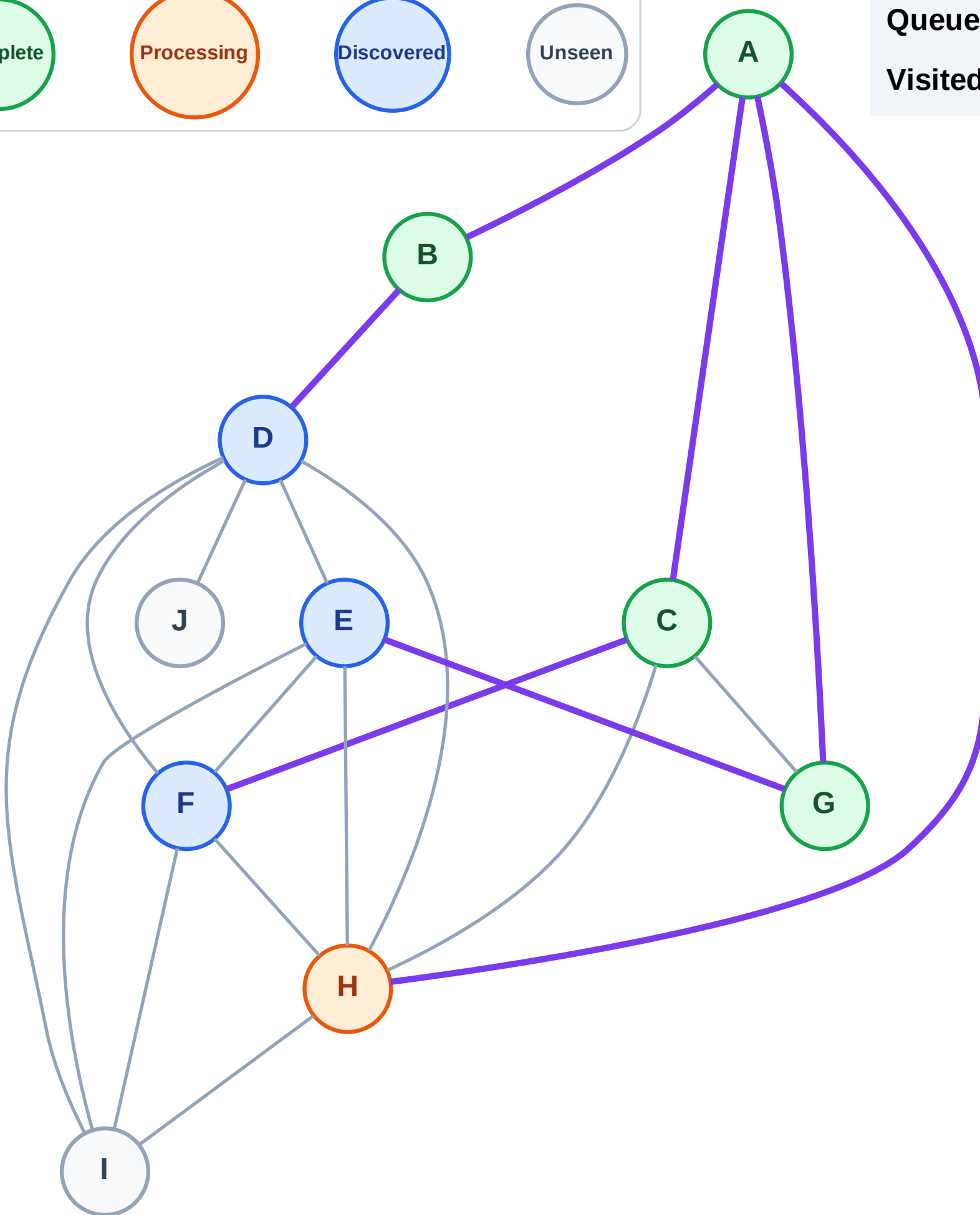
Step 10: Process node H

Legend



Queue: D → F → E

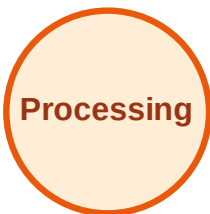
Visited: A, B, C, G



BFS Traversal

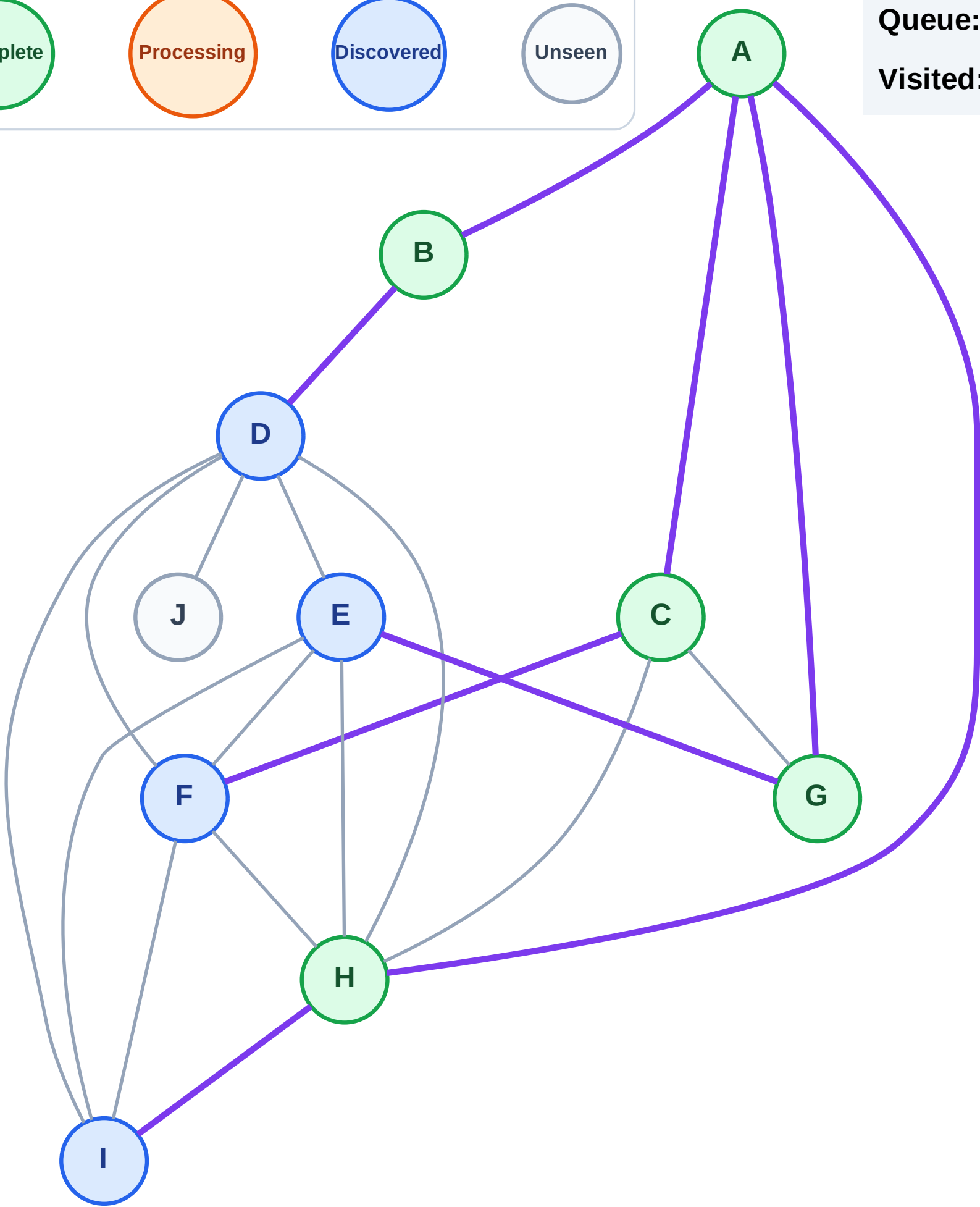
Step 11: Discover I from H

Legend



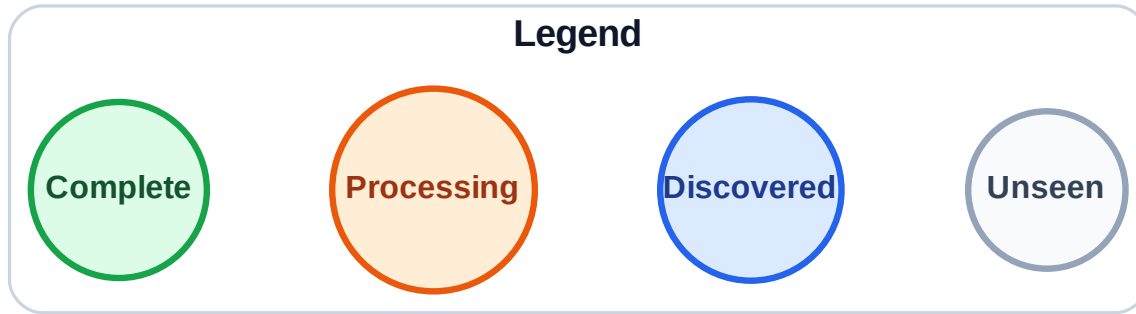
Queue: D → F → E → I

Visited: A, B, C, G, H

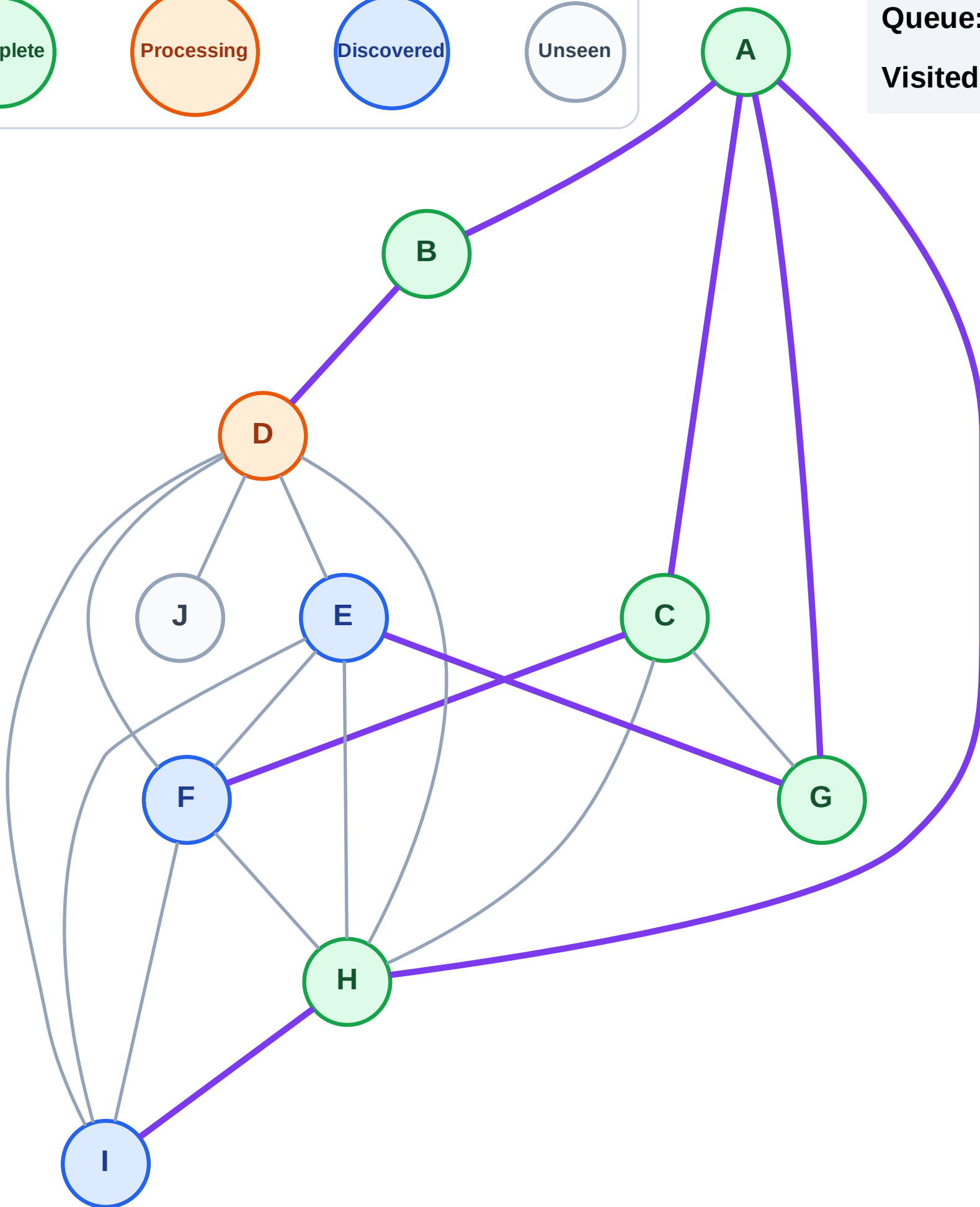


BFS Traversal

Step 12: Process node D



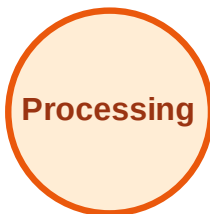
Queue: F → E → I
Visited: A, B, C, G, H



BFS Traversal

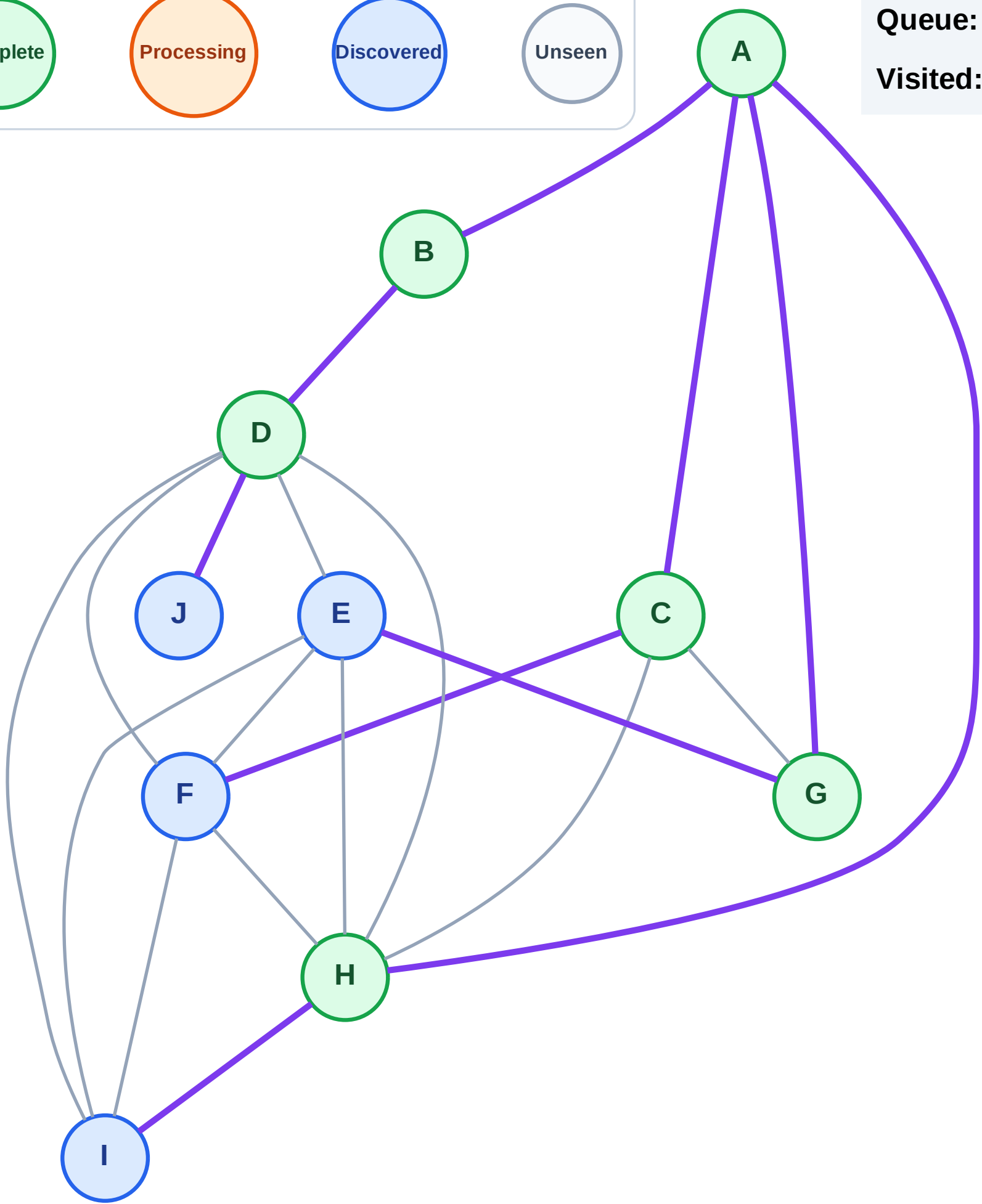
Step 13: Discover J from D

Legend



Queue: F → E → I → J

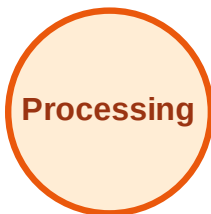
Visited: A, B, C, D, G, H



BFS Traversal

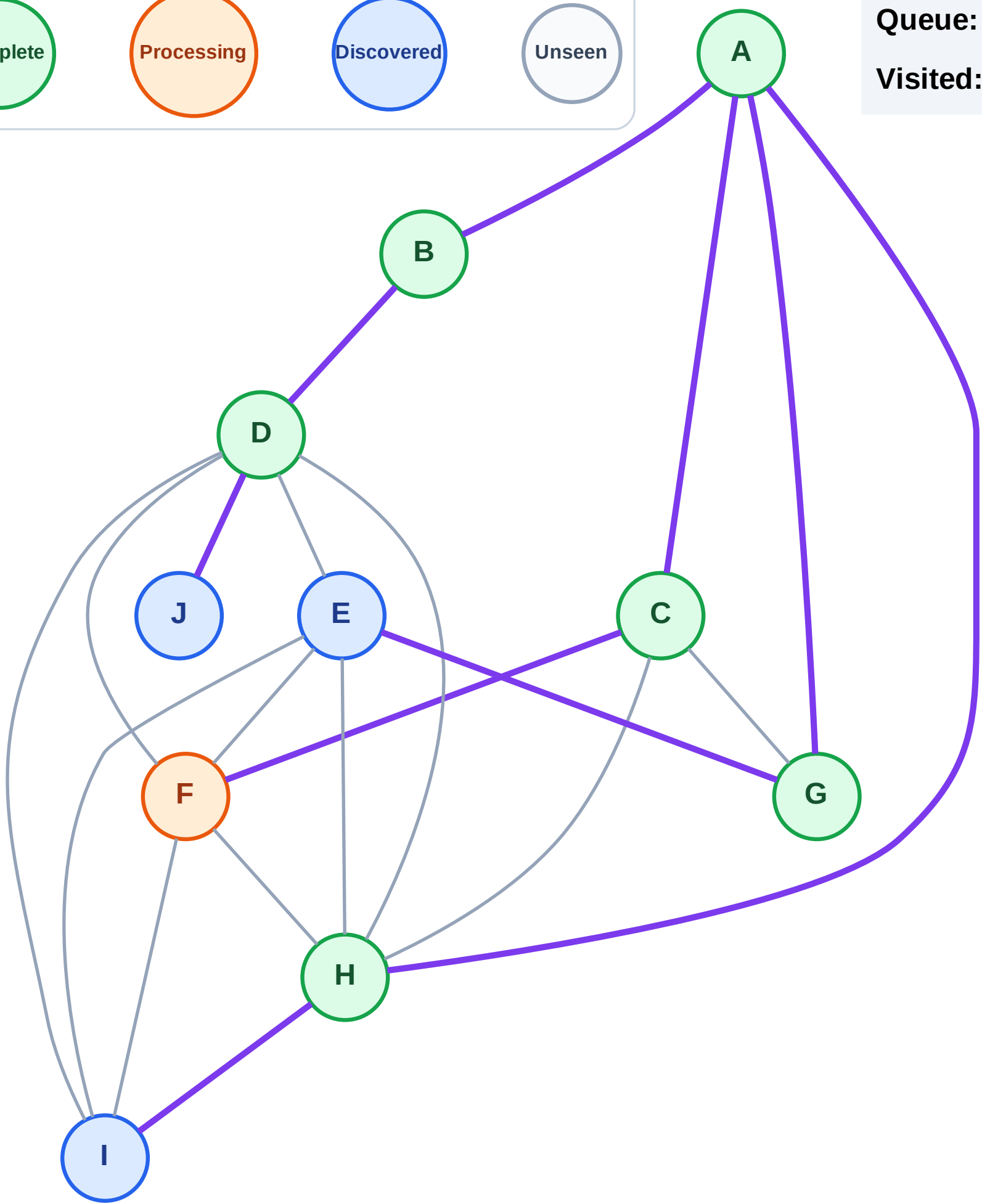
Step 14: Process node F

Legend



Queue: E → I → J

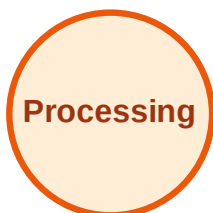
Visited: A, B, C, D, G, H



BFS Traversal

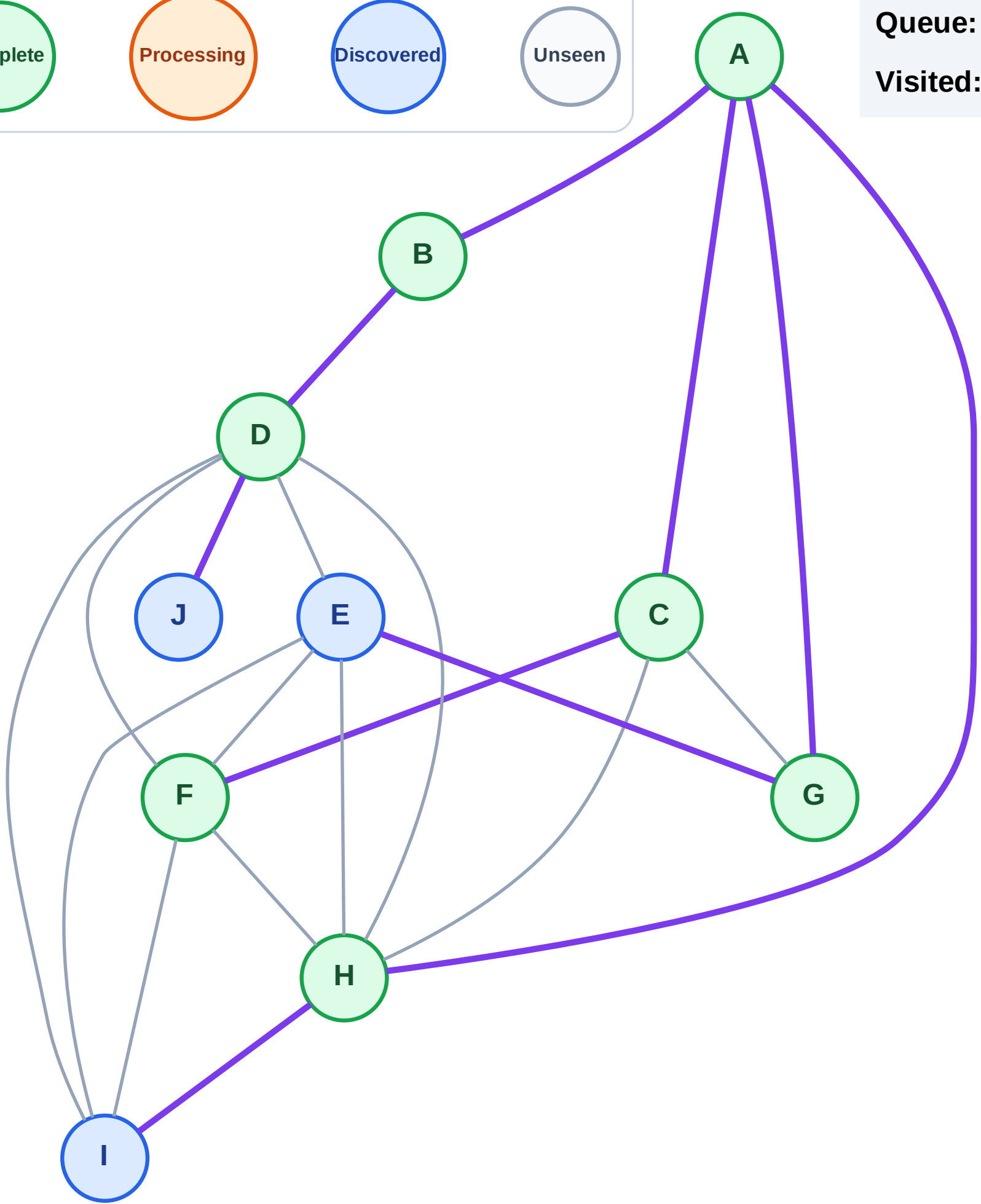
Step 15: No new neighbors from F

Legend



Queue: E → I → J

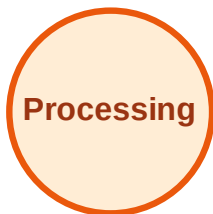
Visited: A, B, C, D, F, G, H



BFS Traversal

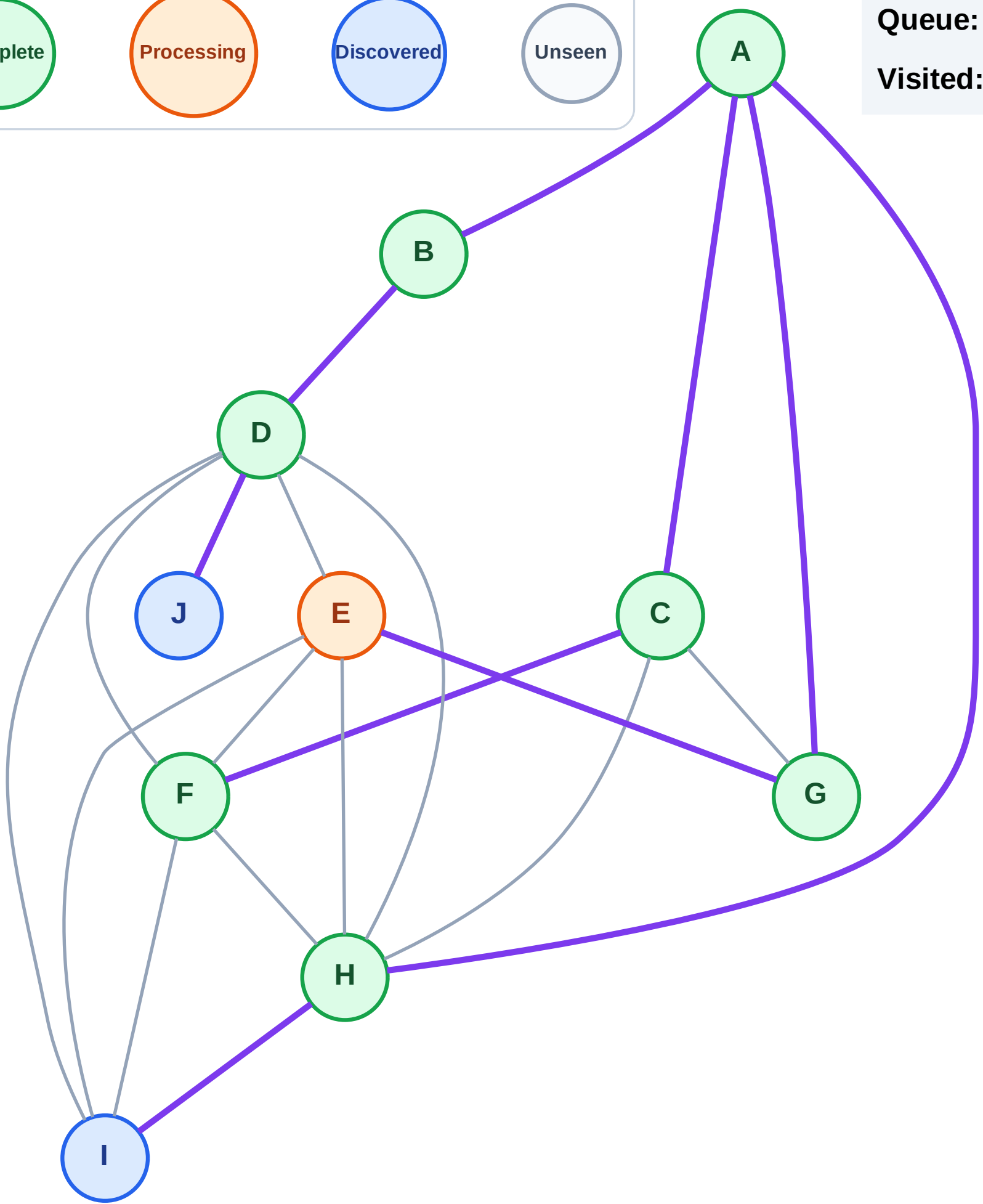
Step 16: Process node E

Legend



Queue: I → J

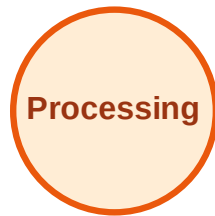
Visited: A, B, C, D, F, G, H



BFS Traversal

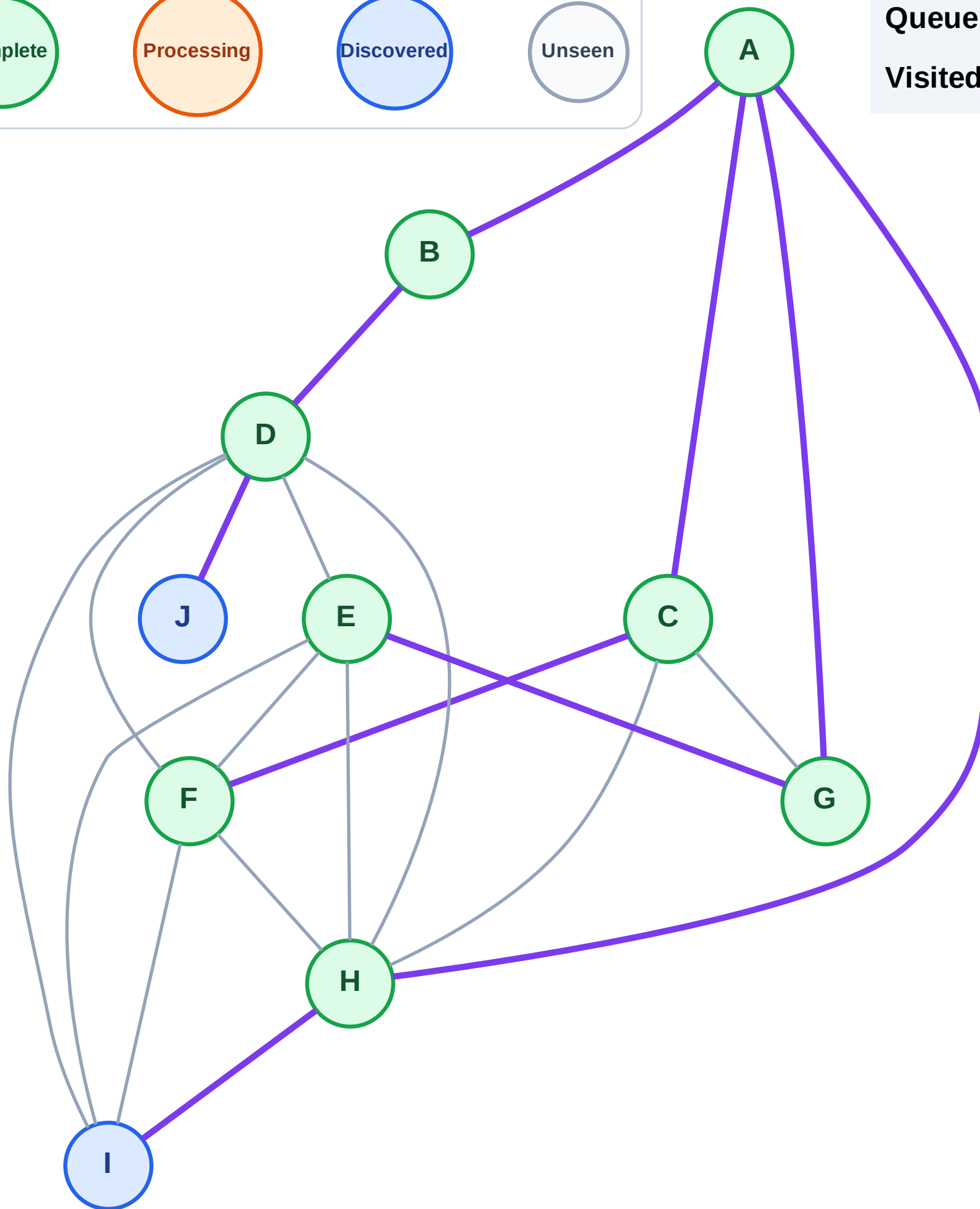
Step 17: No new neighbors from E

Legend



Queue: I → J

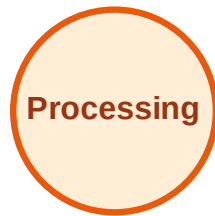
Visited: A, B, C, D, E, F, G, H



BFS Traversal

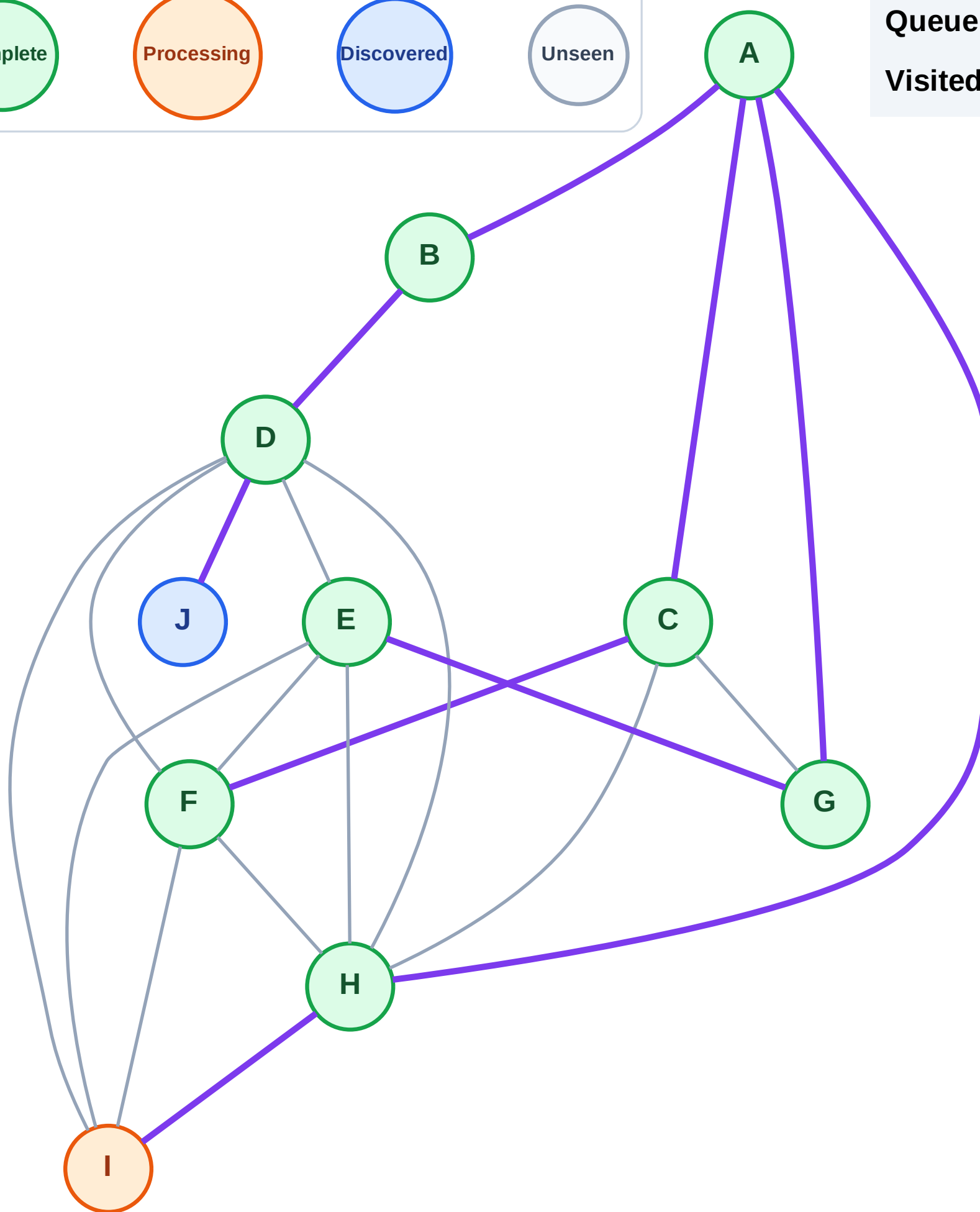
Step 18: Process node I

Legend



Queue: J

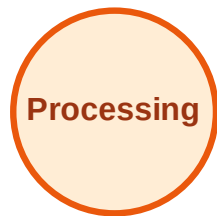
Visited: A, B, C, D, E, F, G, H



BFS Traversal

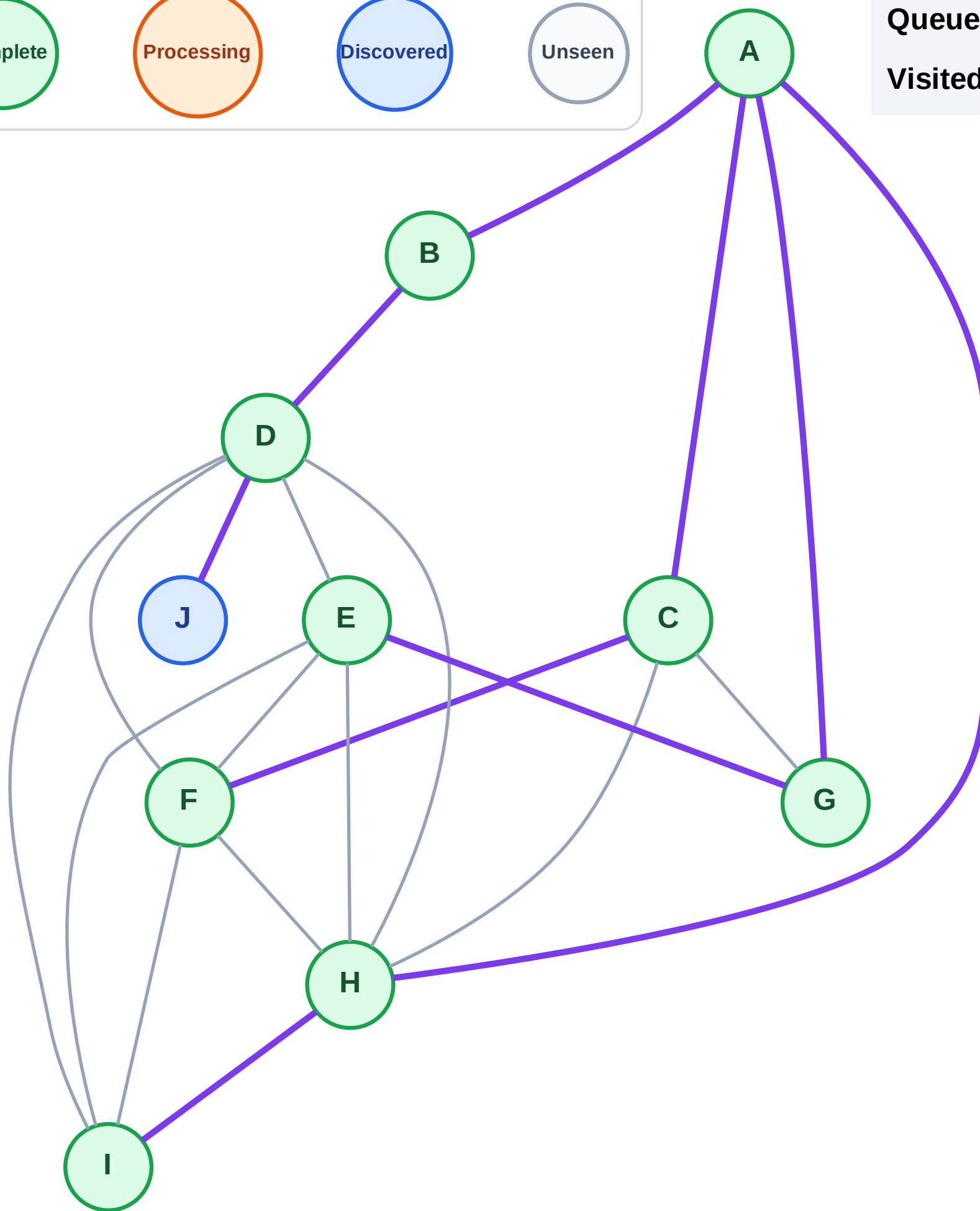
Step 19: No new neighbors from I

Legend



Queue: J

Visited: A, B, C, D, E, F, G, H, I



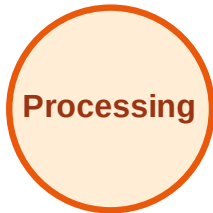
BFS Traversal

Step 20: Process node J

Legend



Complete



Processing



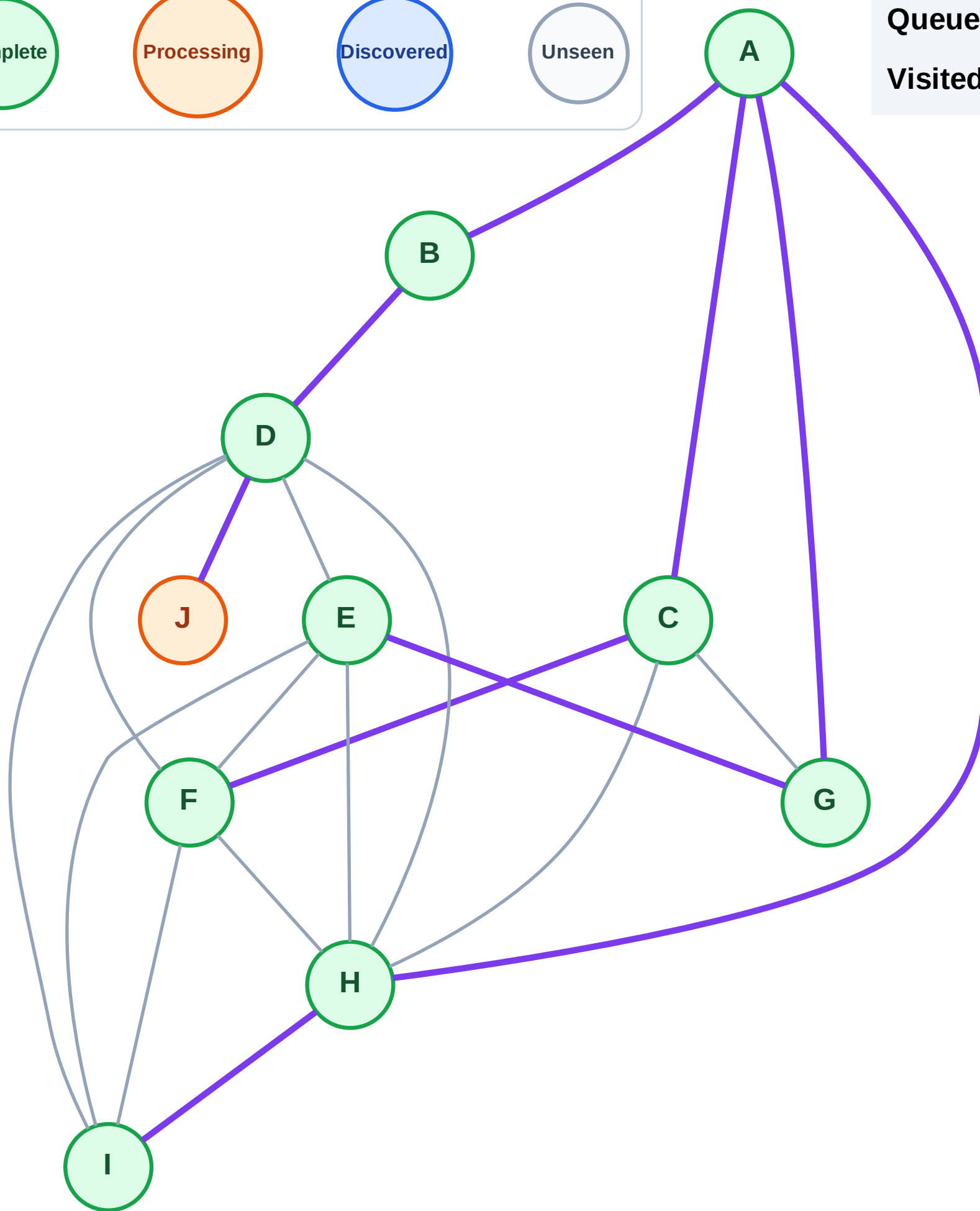
Discovered



Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I

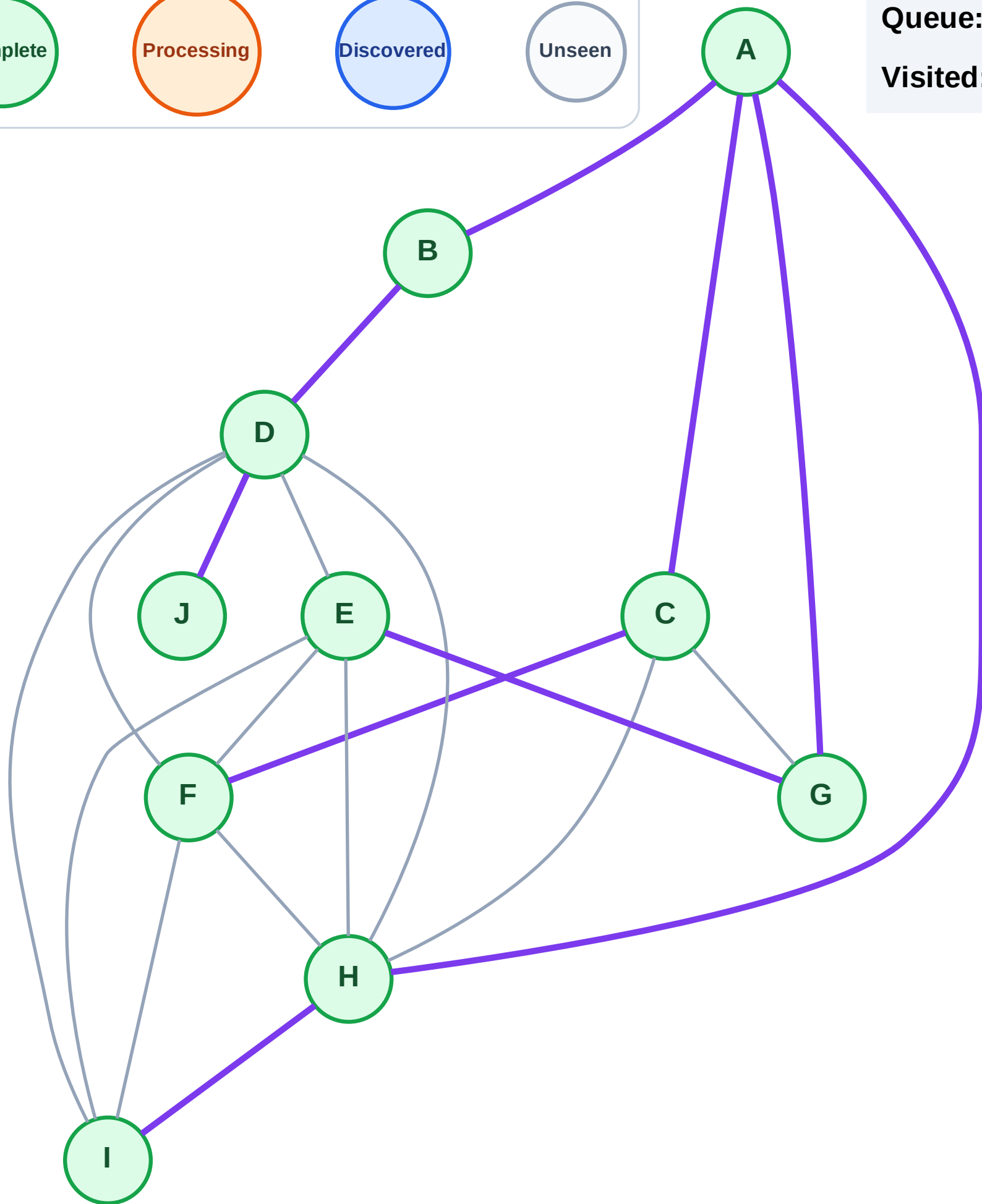


Step 21: No new neighbors from J

Legend



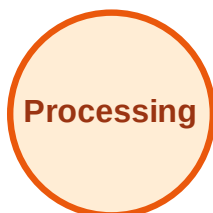
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

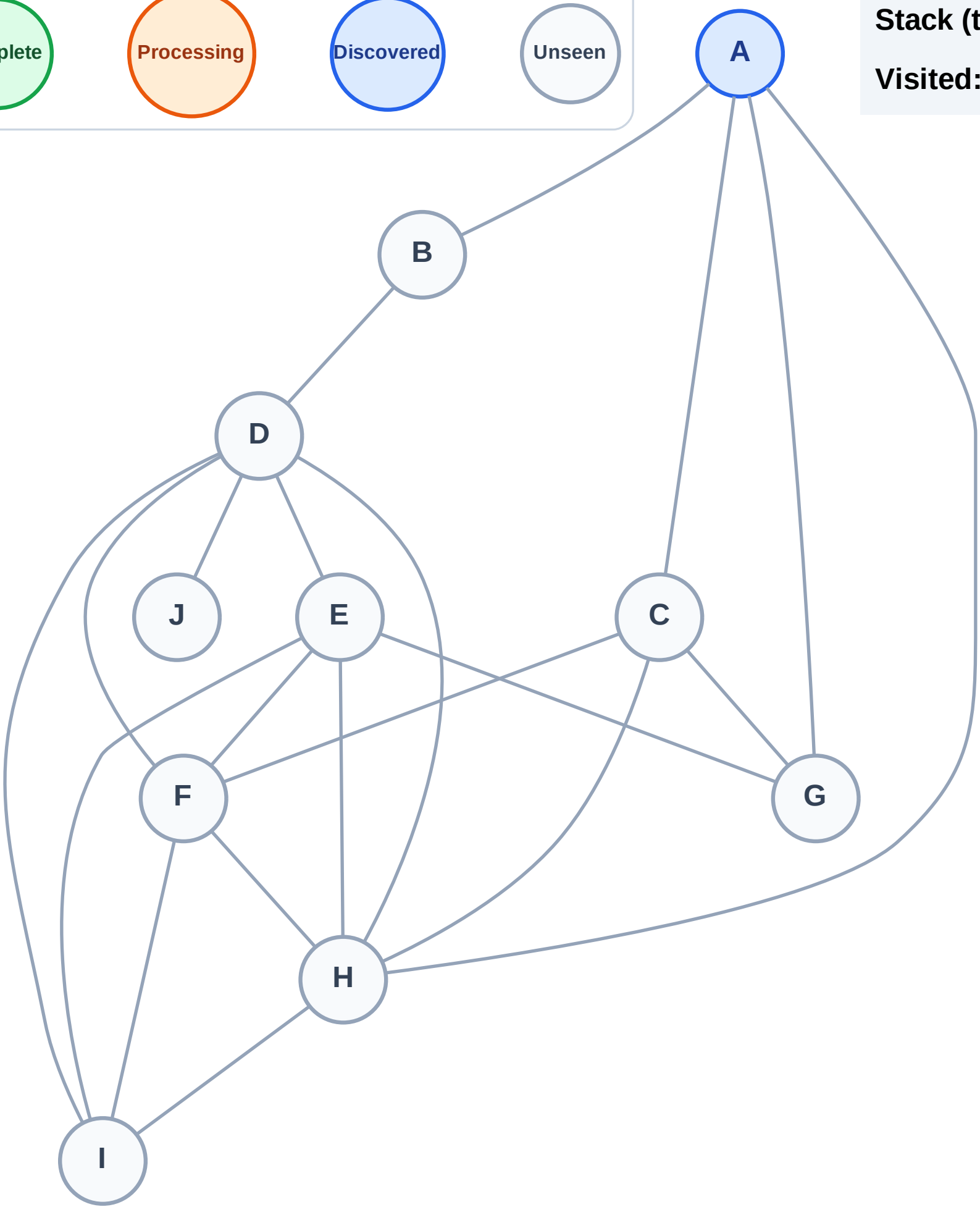
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

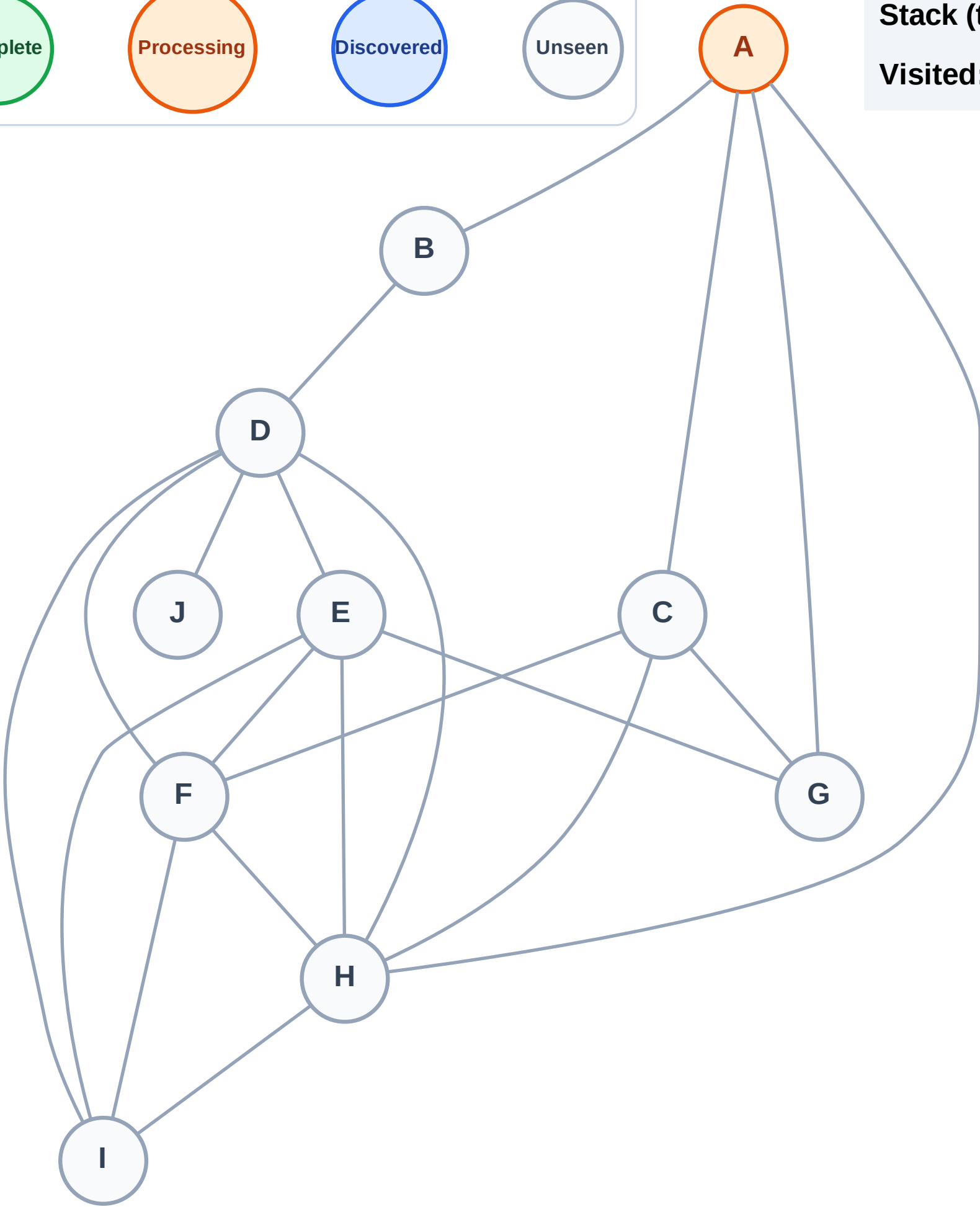
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

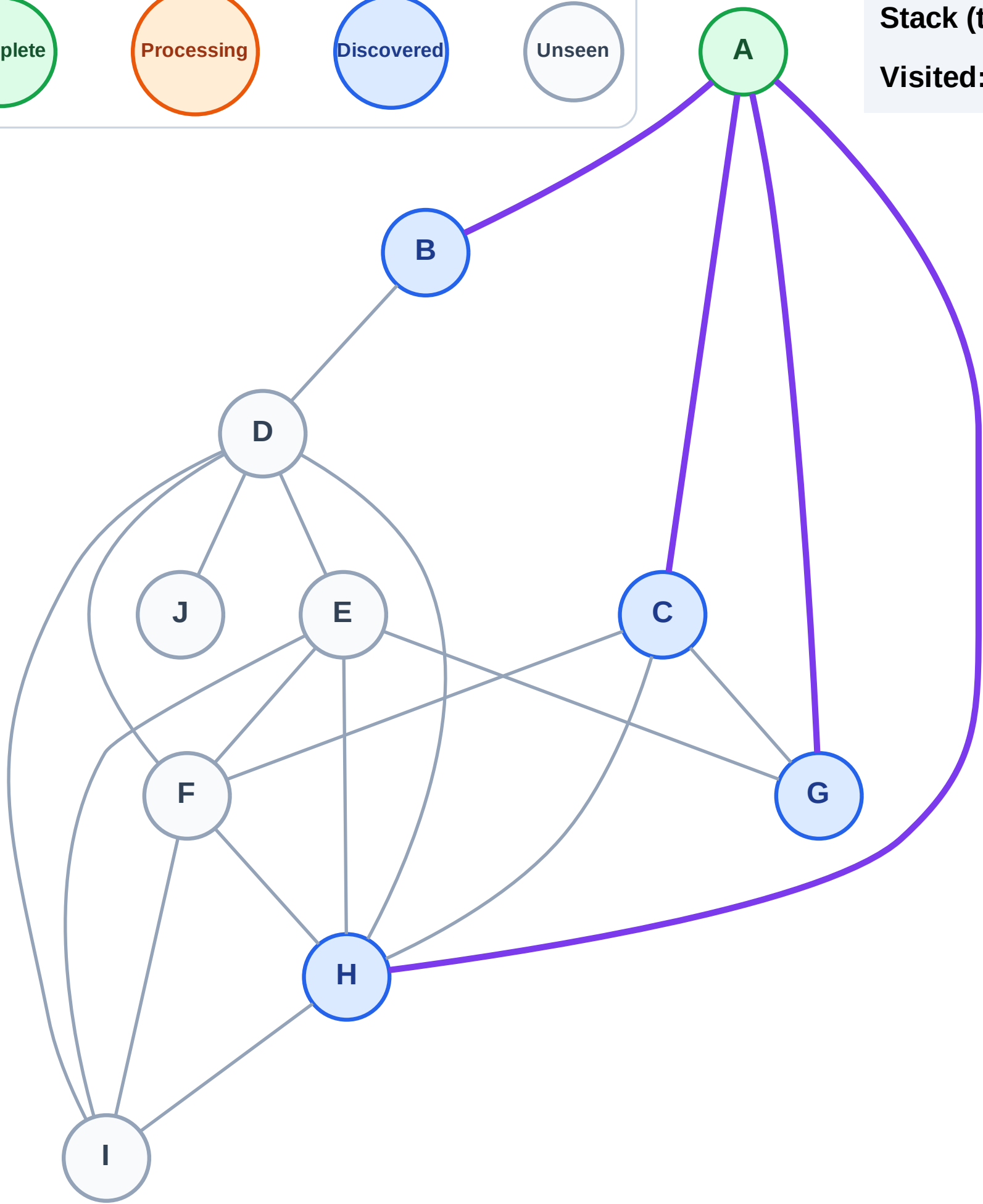
Step 3: Discover H, G, C, B from A

Legend



Stack (top at right): H | G | C | B

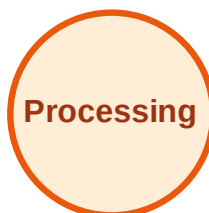
Visited: A



DFS Traversal

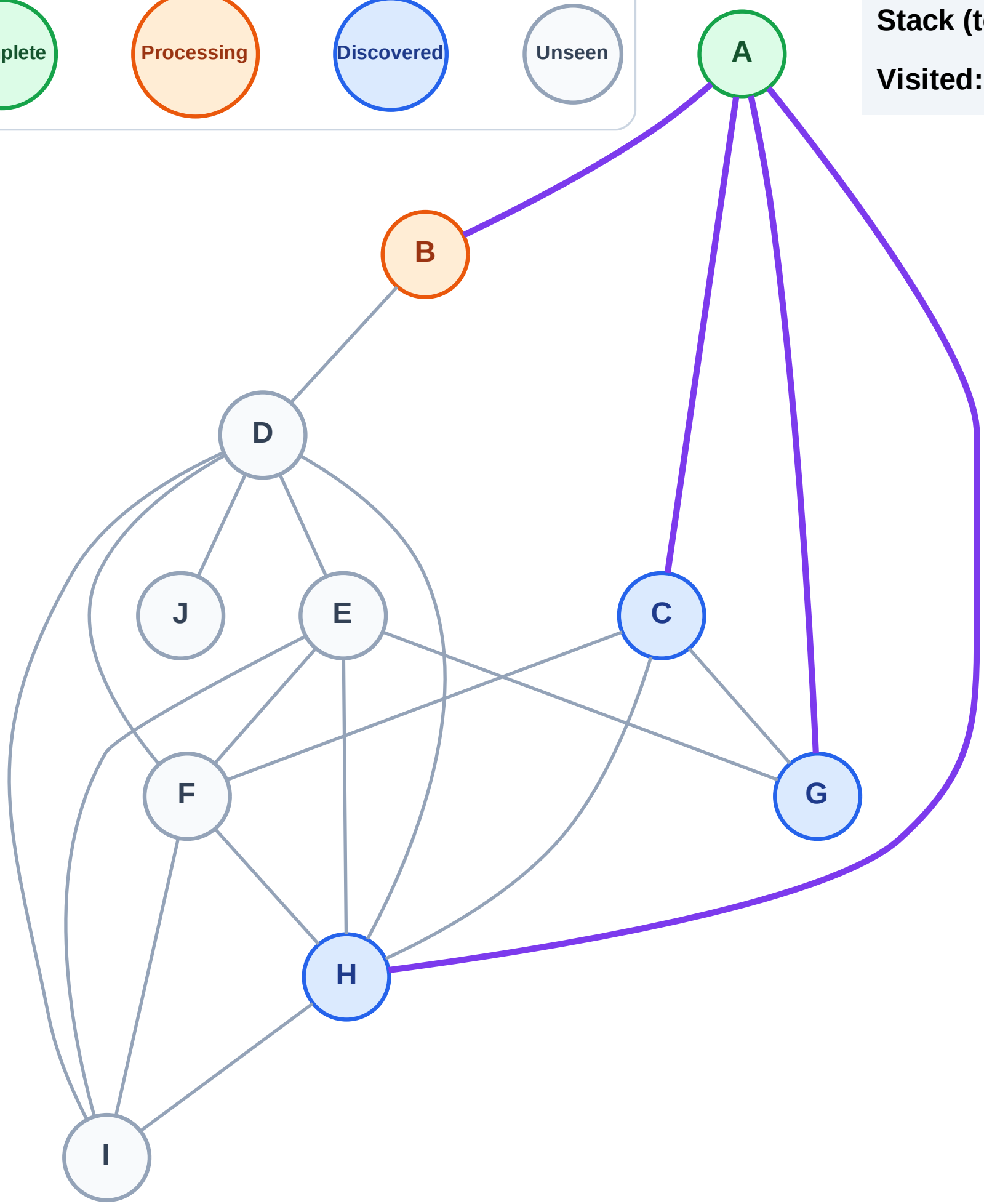
Step 4: Process node B

Legend

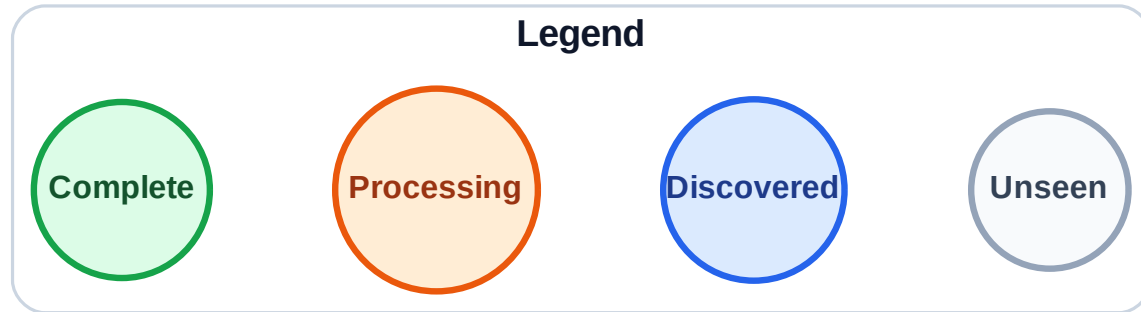


Stack (top at right): H | G | C

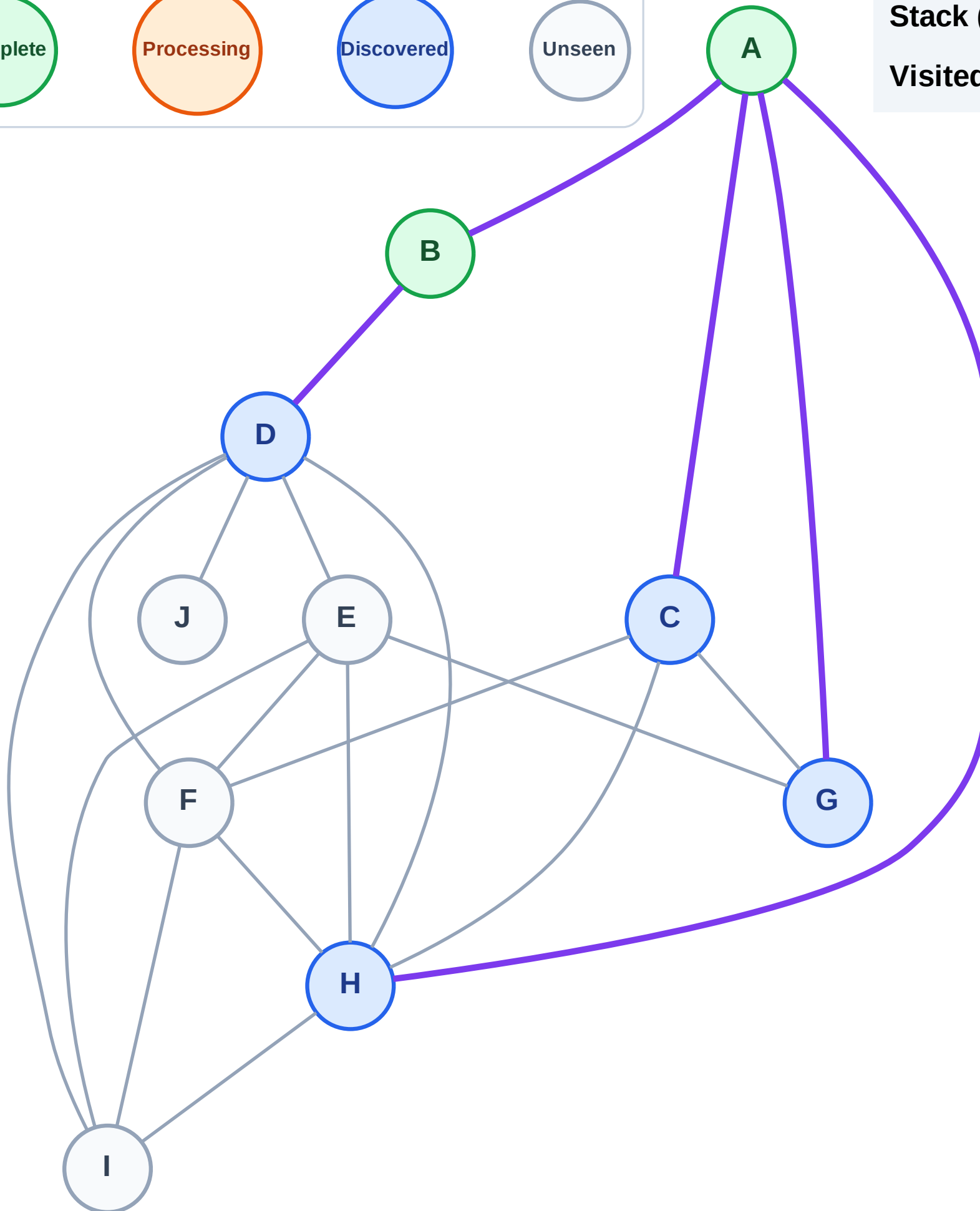
Visited: A



Step 5: Discover D from B



Visited: A, B



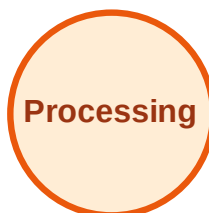
DFS Traversal

Step 6: Process node D

Legend



Complete



Processing



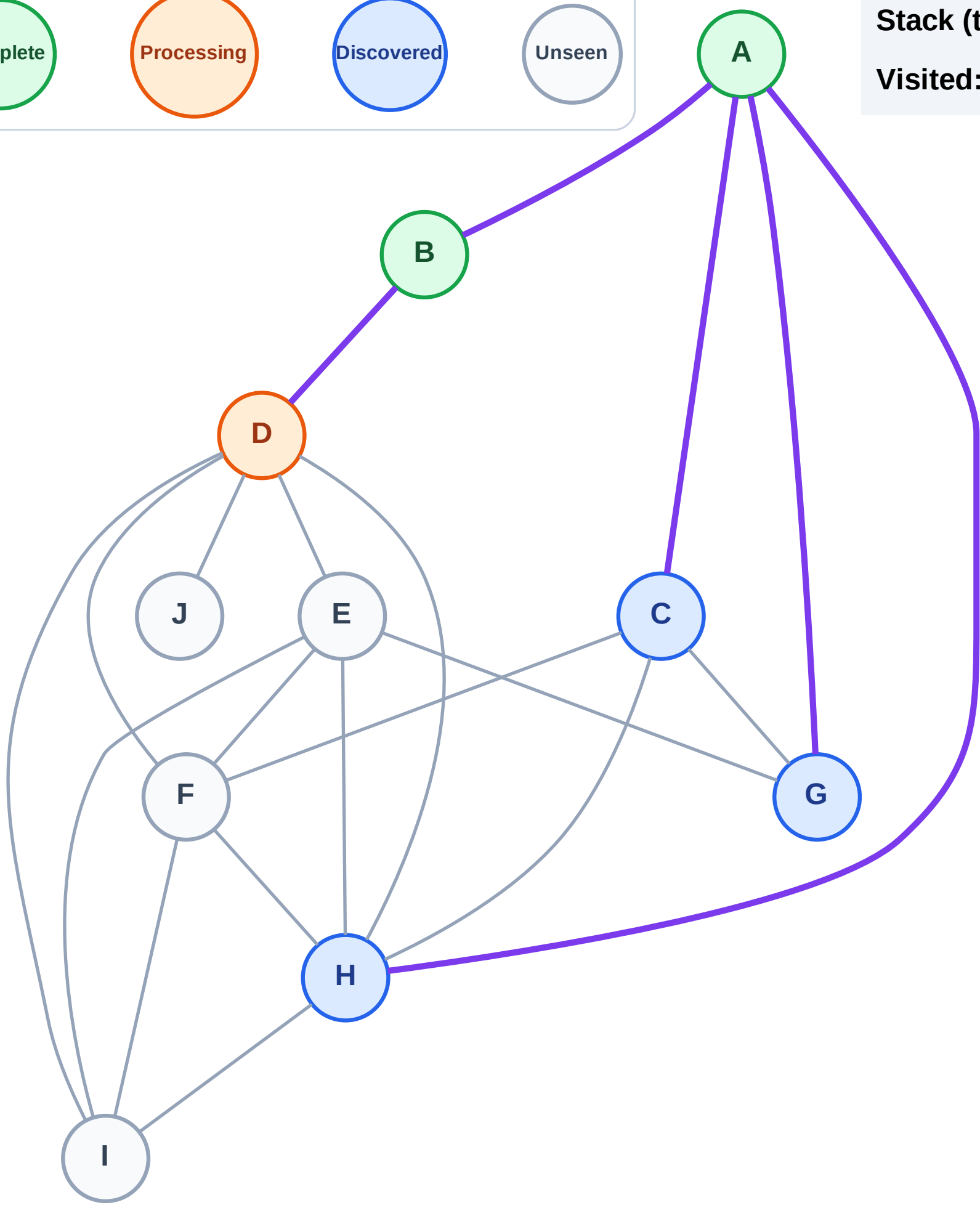
Discovered



Unseen

Stack (top at right): H | G | C

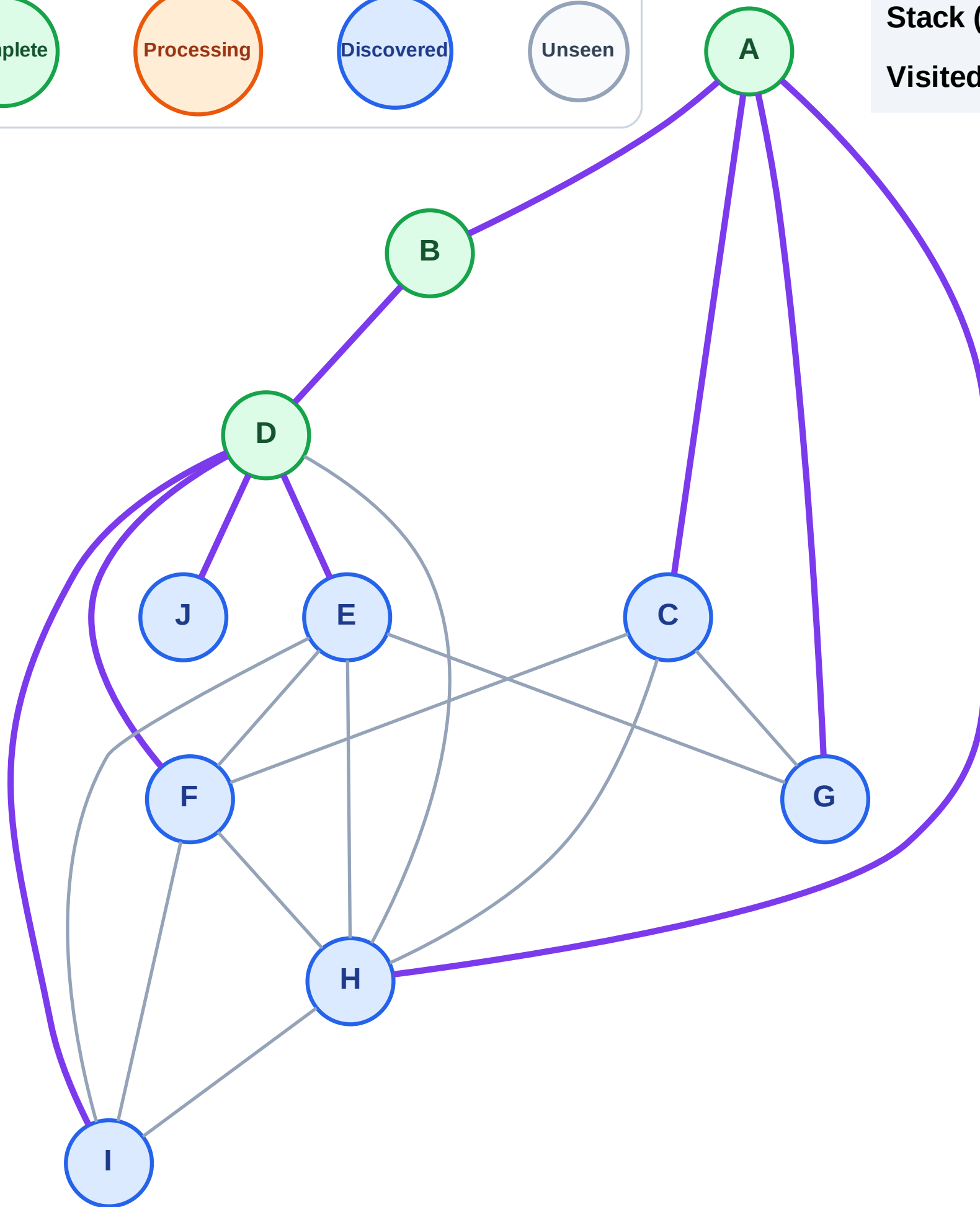
Visited: A, B



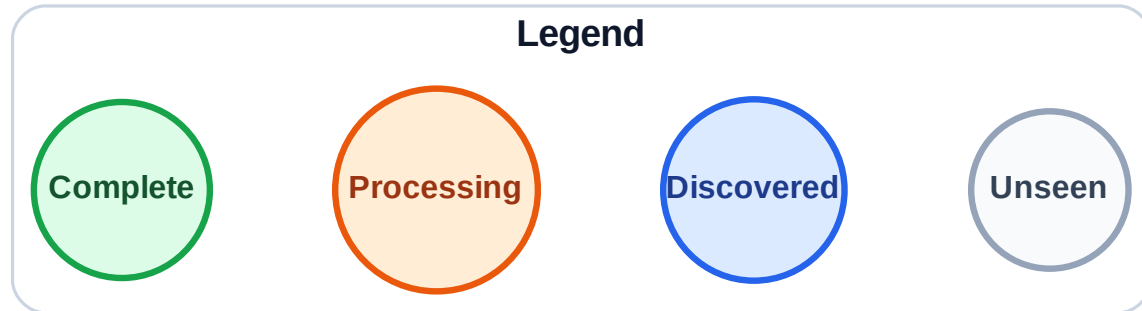
Step 7: Discover J, I, F, E from D

Stack (top at right): H | G | C | J | I | F | E

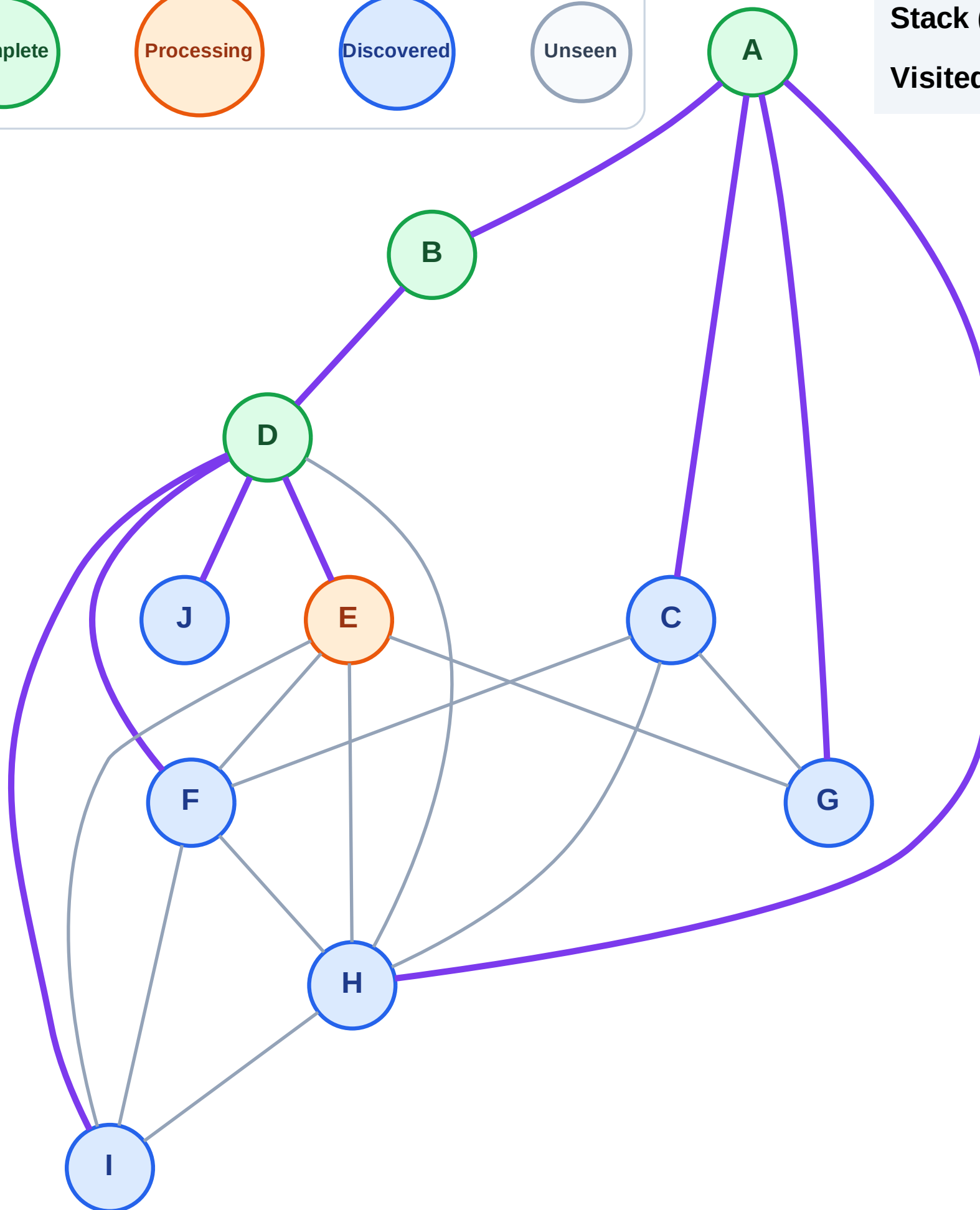
Visited: A, B, D



Step 8: Process node E



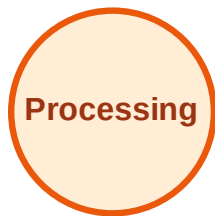
Visited: A, B, D



DFS Traversal

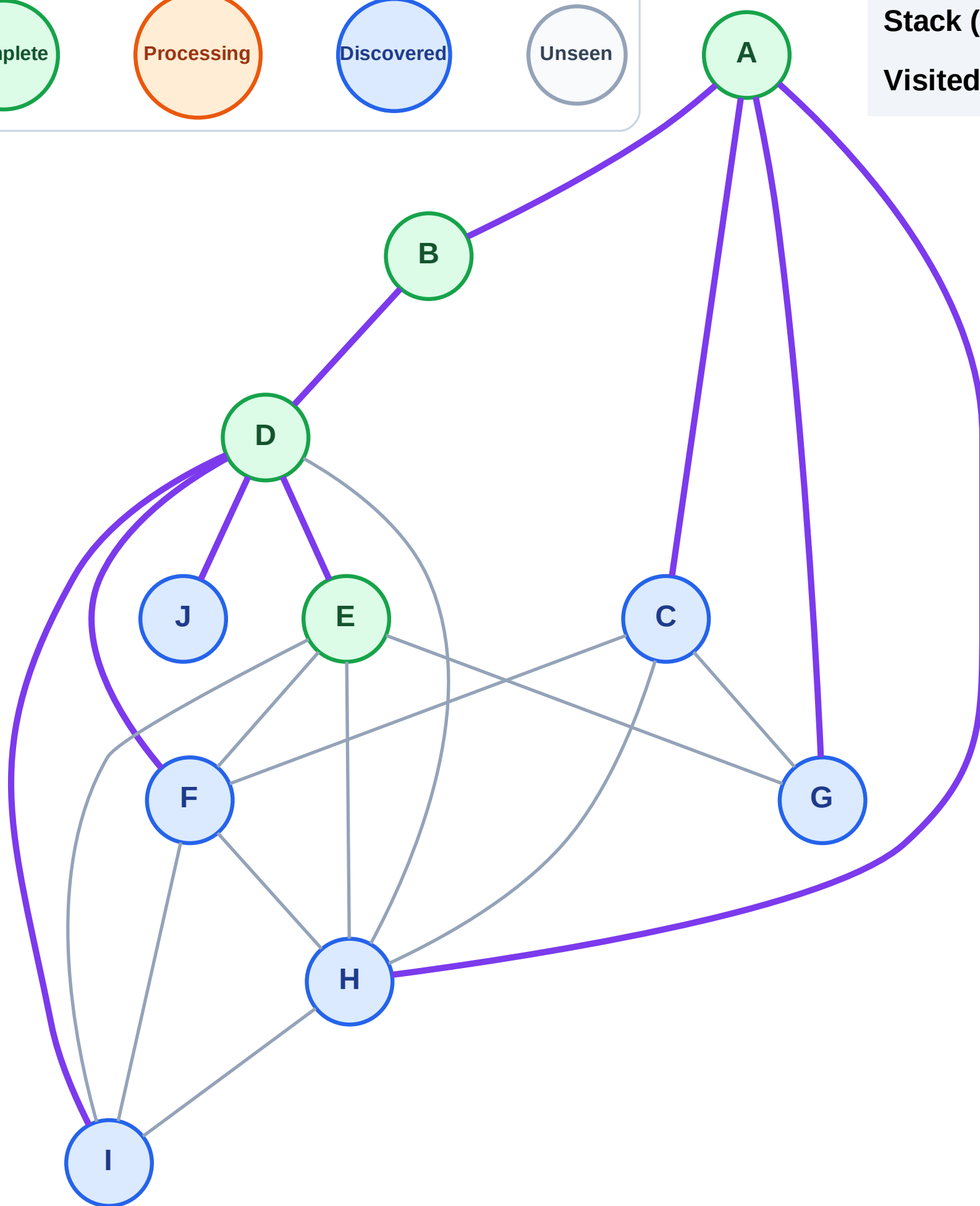
Step 9: No new neighbors from E

Legend



Stack (top at right): H | G | C | J | I | F

Visited: A, B, D, E



DFS Traversal

Step 10: Process node F

Legend

Complete

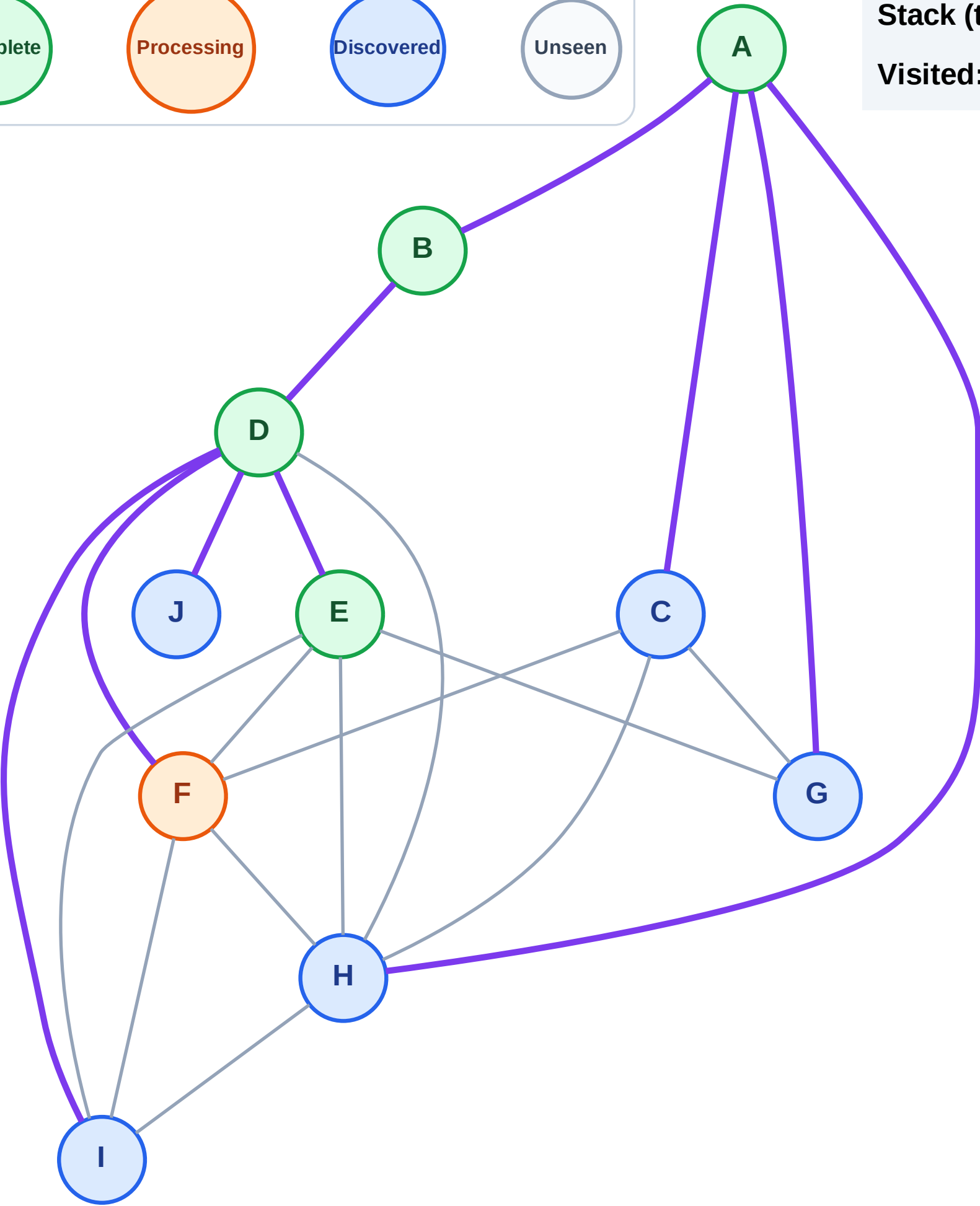
Processing

Discovered

Unseen

Stack (top at right): H | G | C | J | I

Visited: A, B, D, E



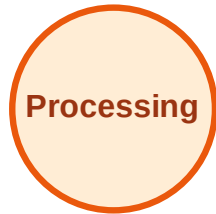
DFS Traversal

Step 11: No new neighbors from F

Legend



Complete



Processing



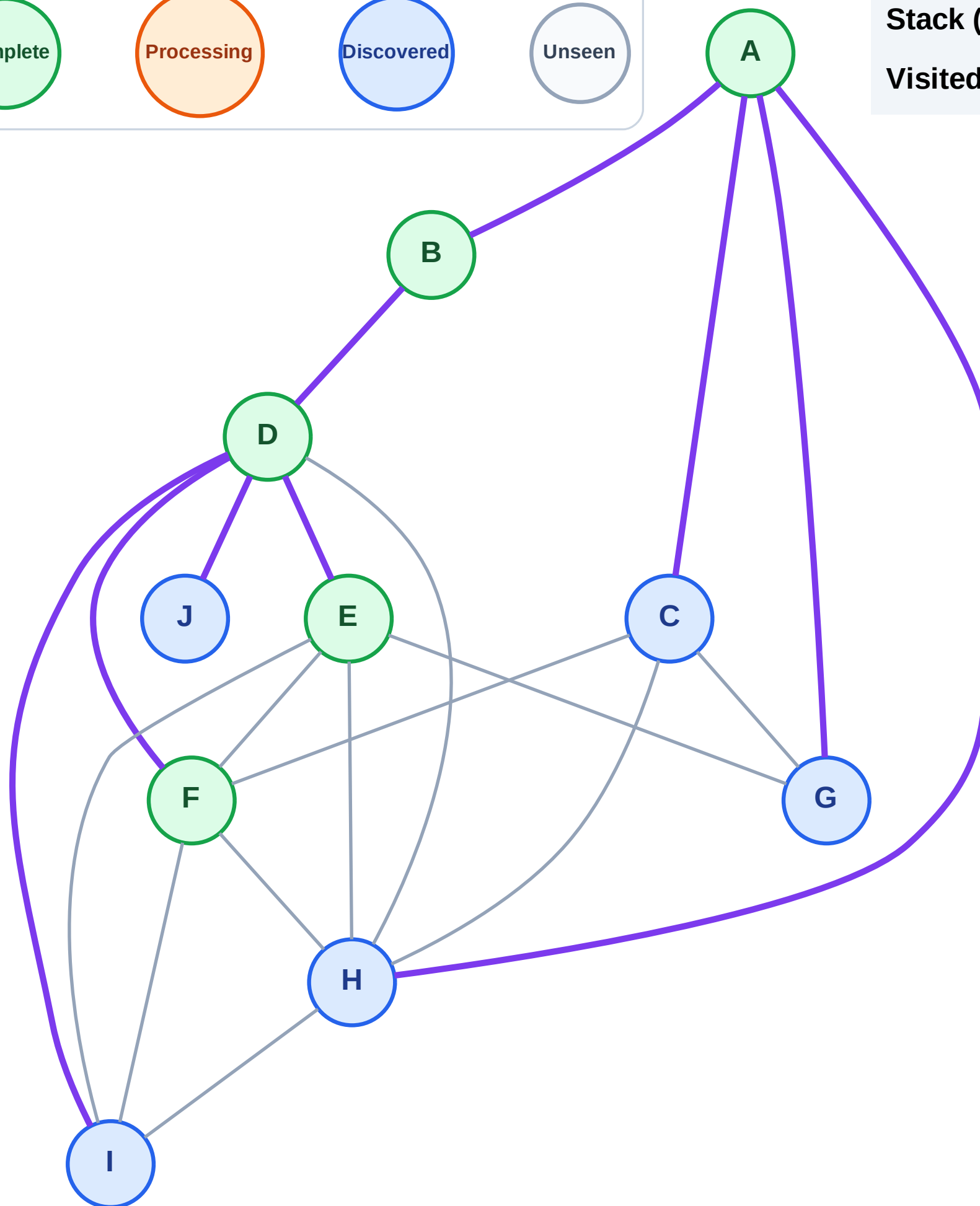
Discovered



Unseen

Stack (top at right): H | G | C | J | I

Visited: A, B, D, E, F



DFS Traversal

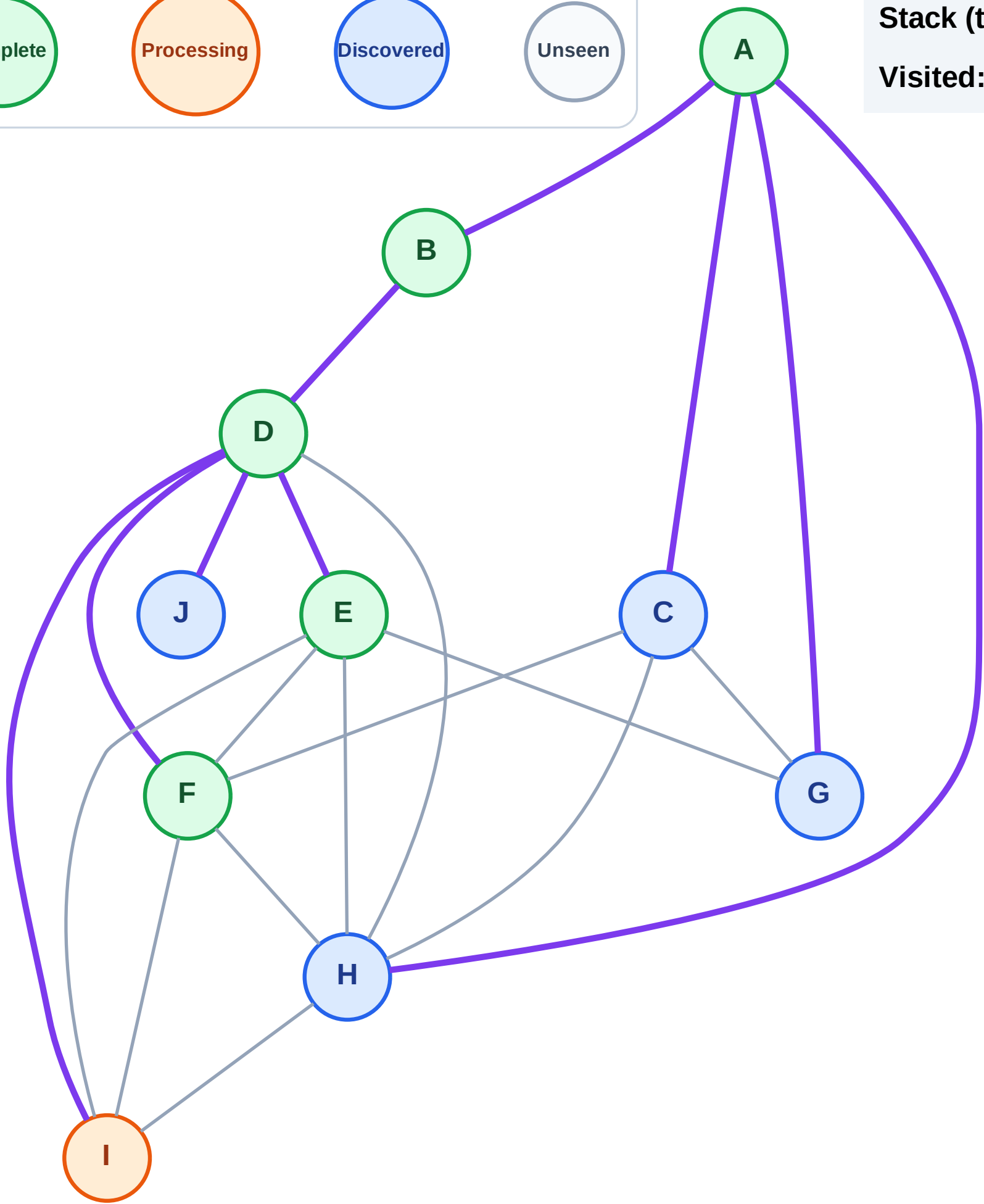
Step 12: Process node I

Legend



Stack (top at right): H | G | C | J

Visited: A, B, D, E, F



DFS Traversal

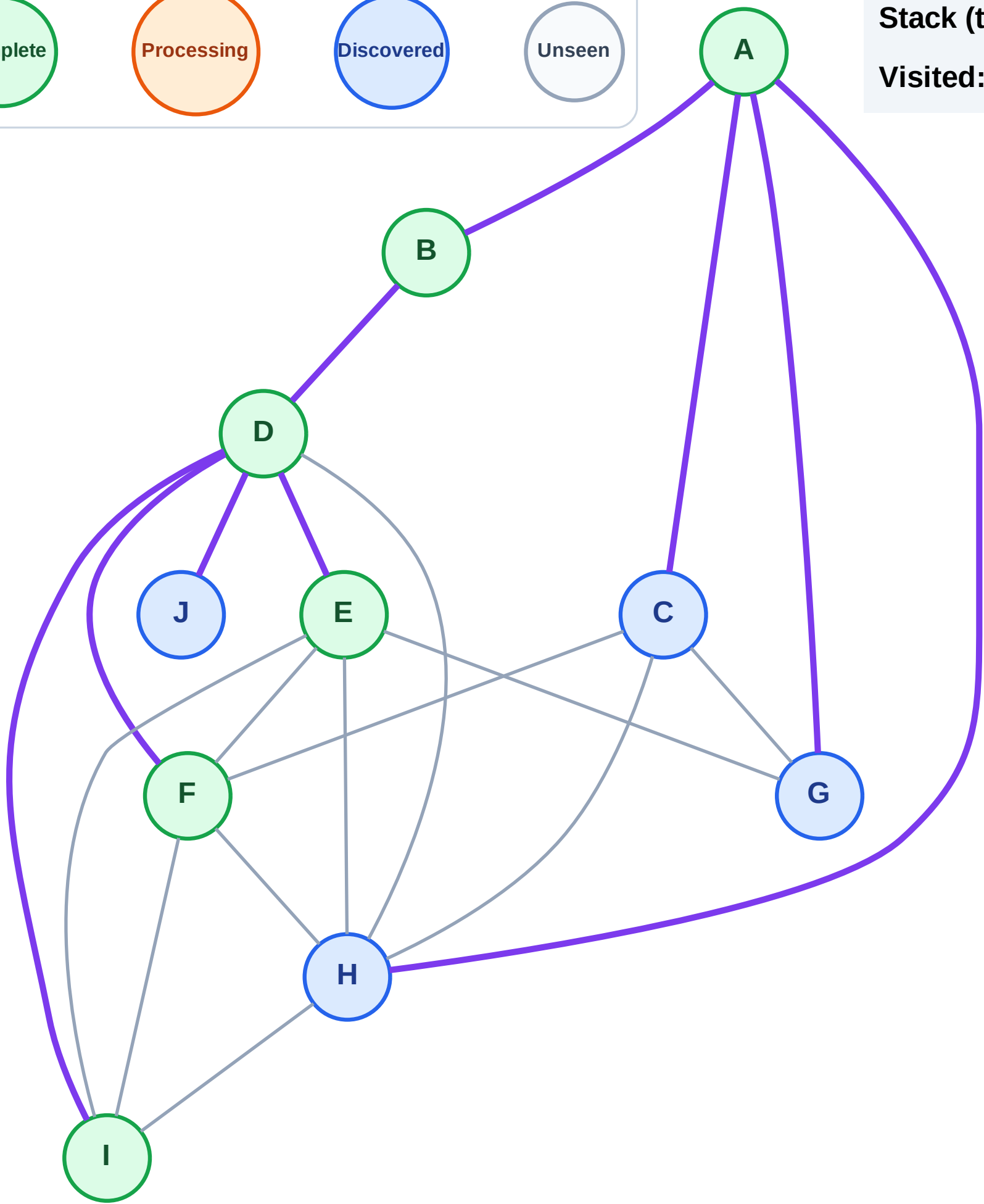
Step 13: No new neighbors from I

Legend

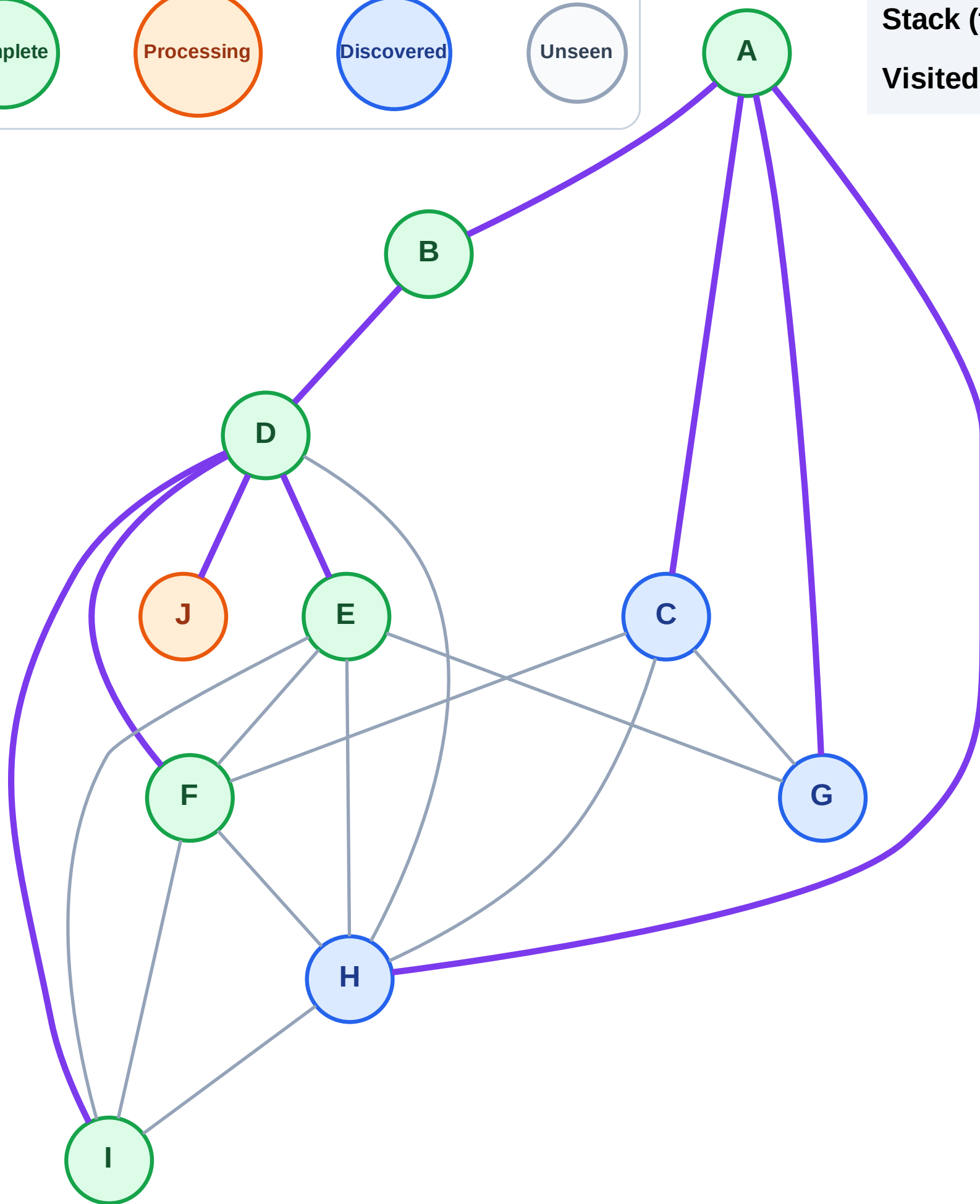
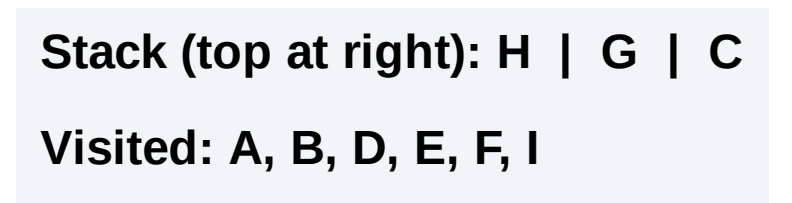


Stack (top at right): H | G | C | J

Visited: A, B, D, E, F, I



Step 14: Process node J



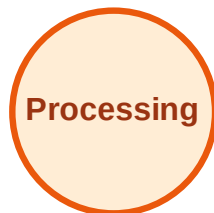
DFS Traversal

Step 15: No new neighbors from J

Legend



Complete



Processing



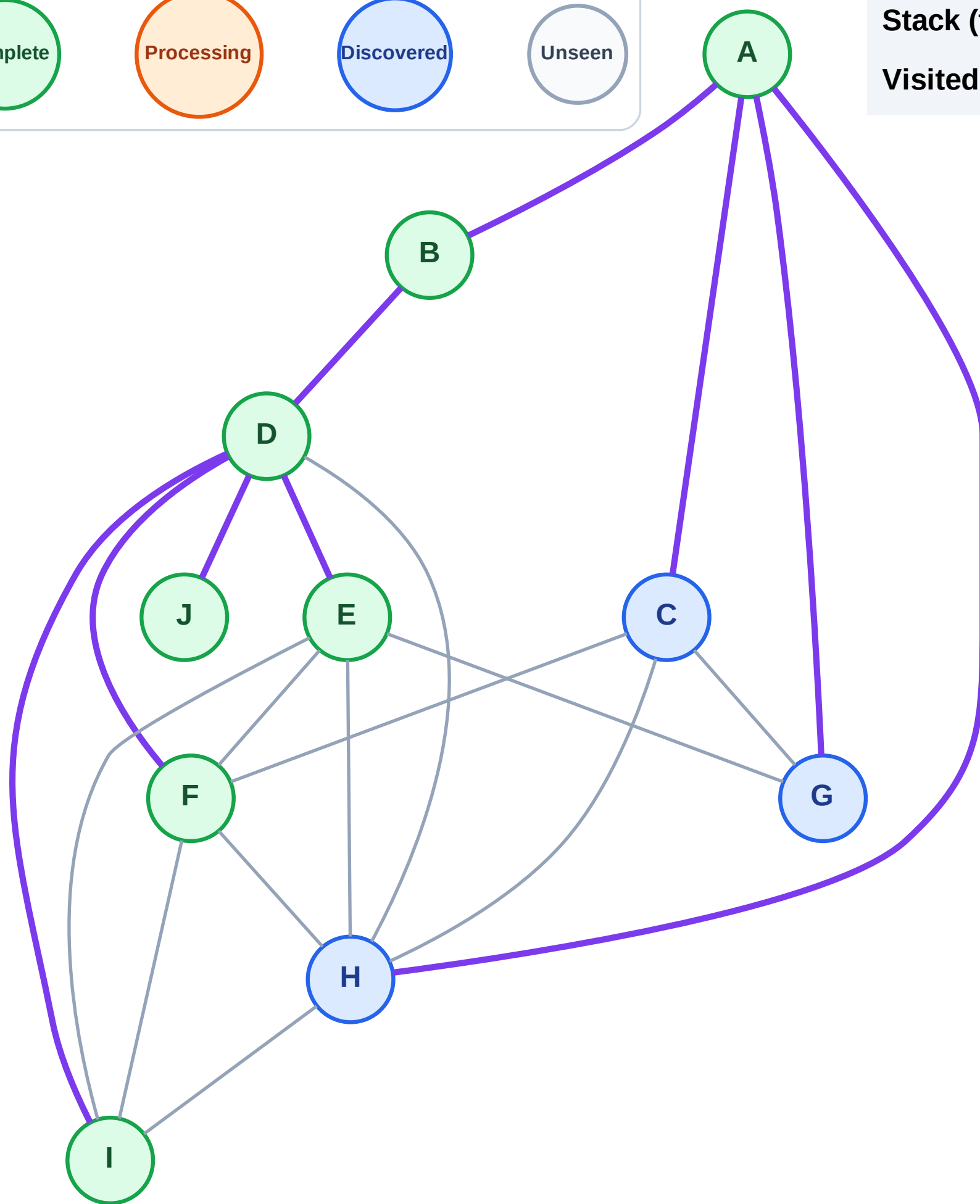
Discovered



Unseen

Stack (top at right): H | G | C

Visited: A, B, D, E, F, I, J



DFS Traversal

Step 16: Process node C

Legend

Complete

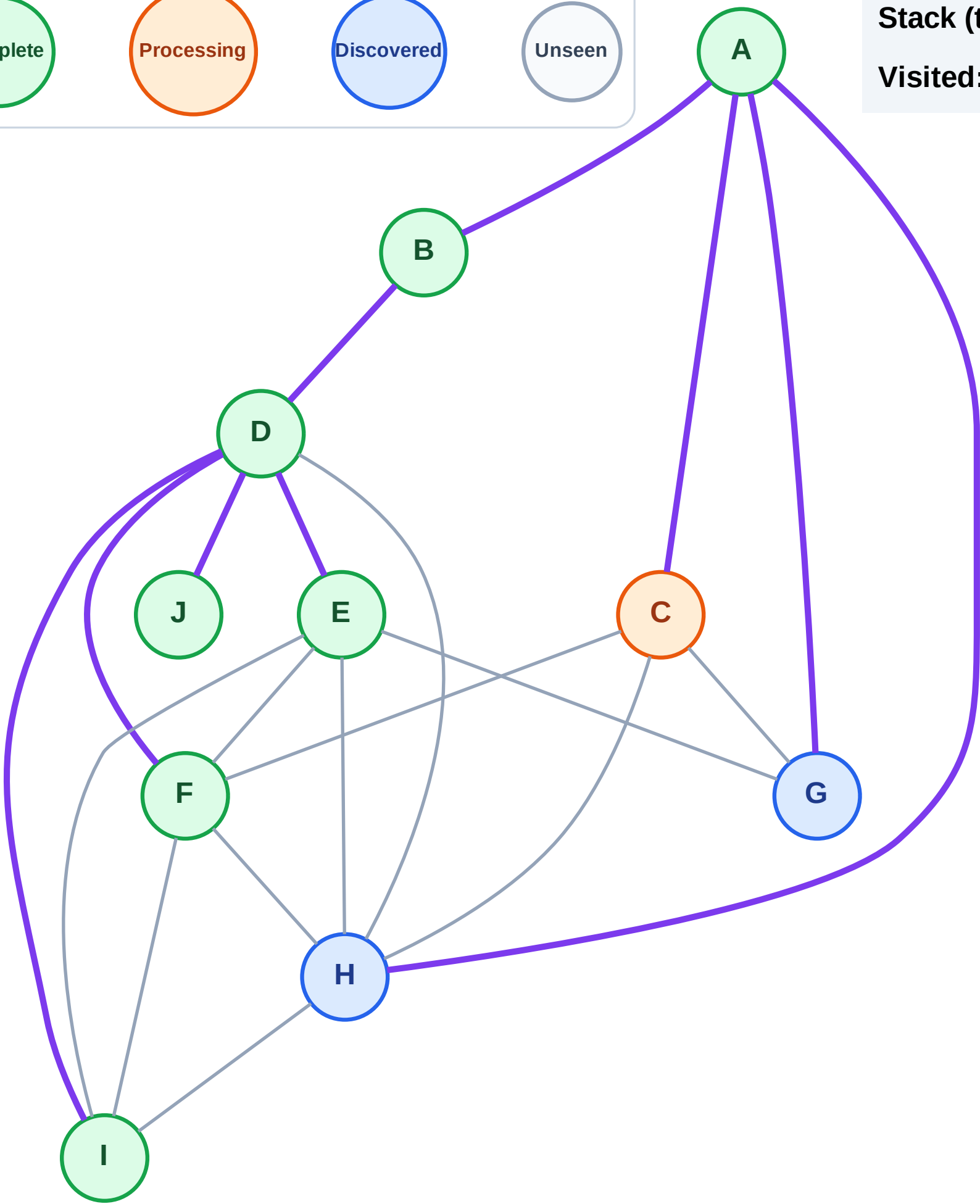
Processing

Discovered

Unseen

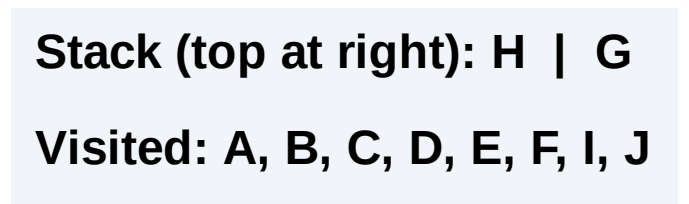
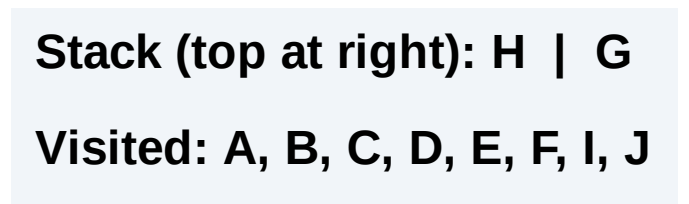
Stack (top at right): H | G

Visited: A, B, D, E, F, I, J



Step 17: No new neighbors from C

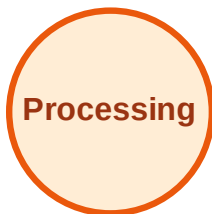
Step 17: No new neighbors from C



DFS Traversal

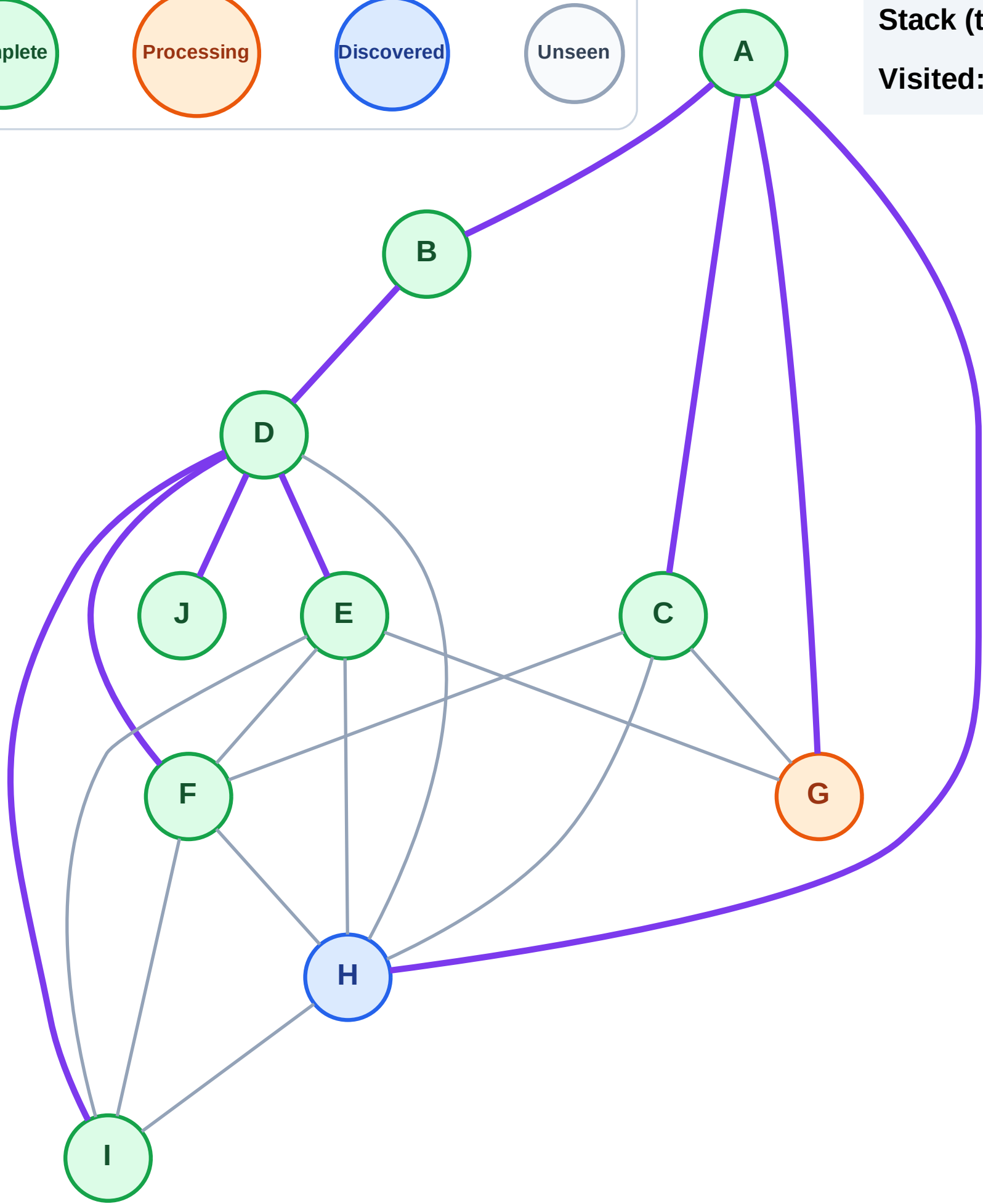
Step 18: Process node G

Legend



Stack (top at right): H

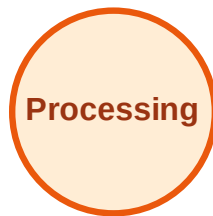
Visited: A, B, C, D, E, F, I, J



DFS Traversal

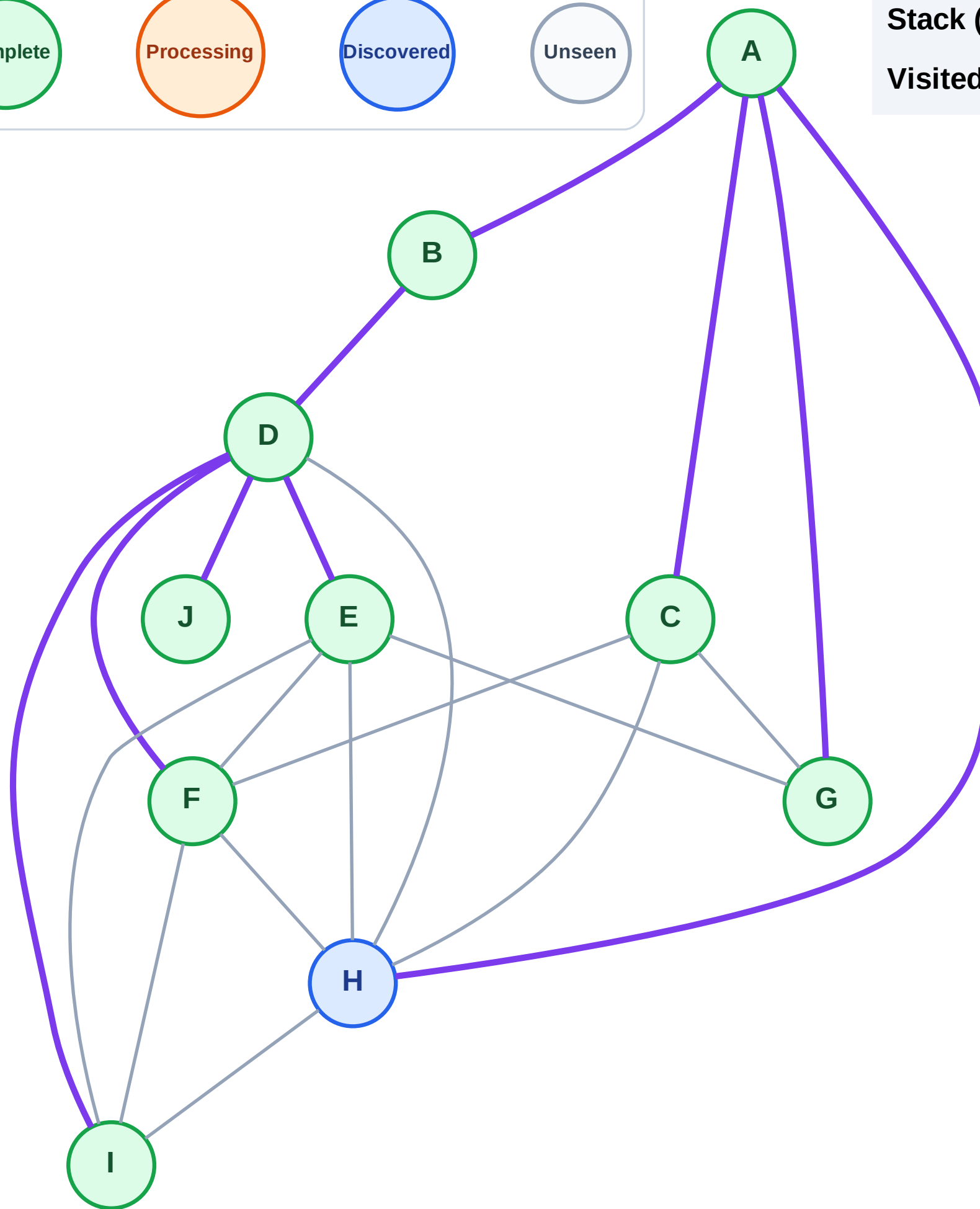
Step 19: No new neighbors from G

Legend

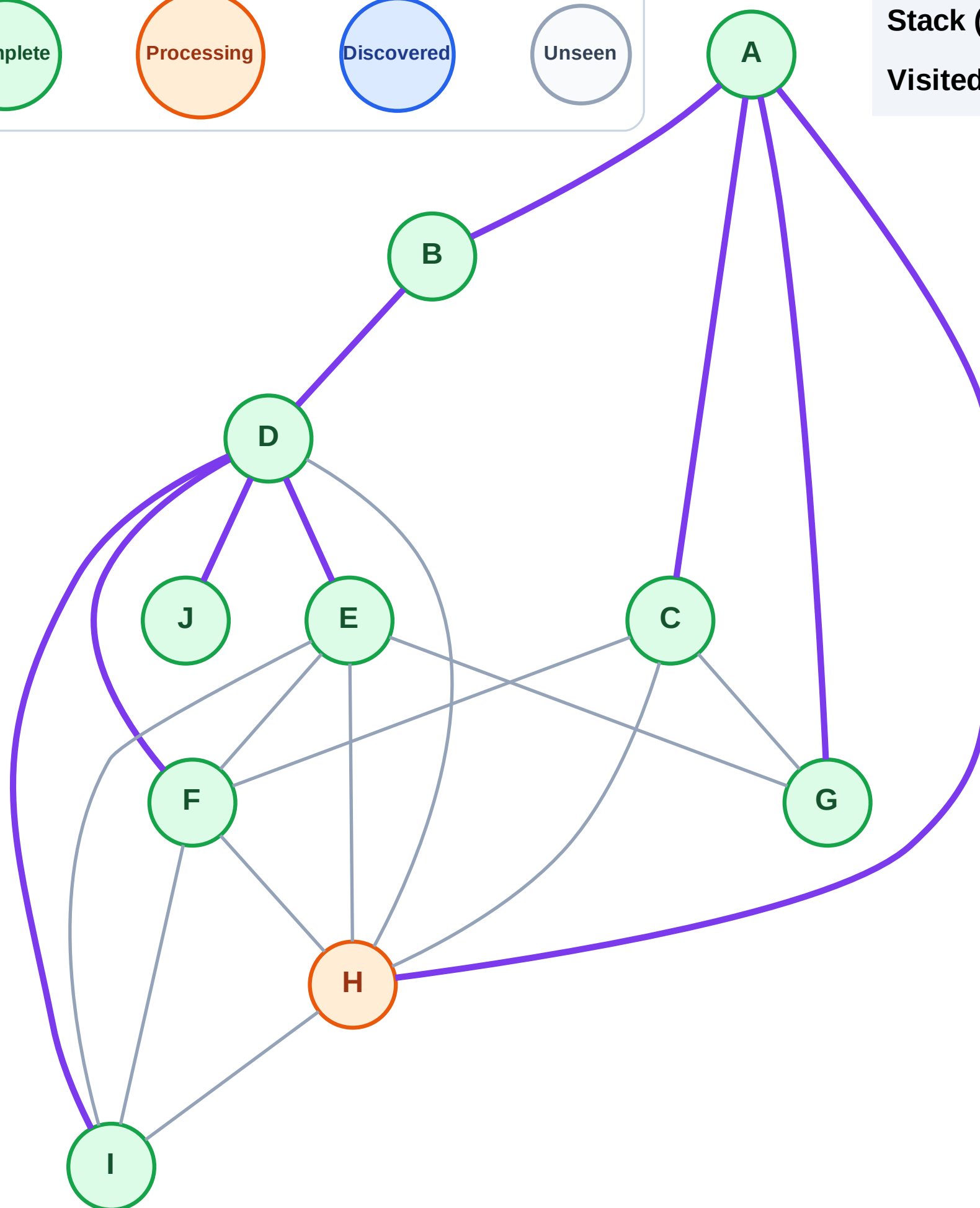
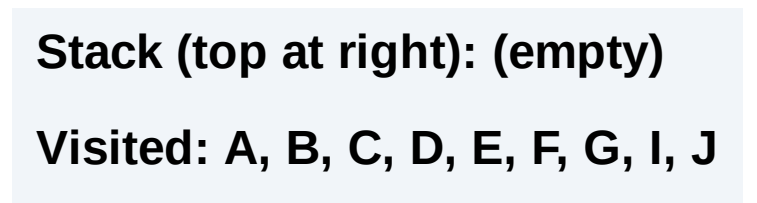


Stack (top at right): H

Visited: A, B, C, D, E, F, G, I, J



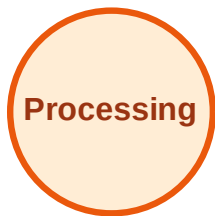
Step 20: Process node H



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

